HANDSMEN THREADS

Elevating the Art of Sophistication in Men's Fashion

Salesforce Certified Platform Developer Project Report

Prepared with Salesforce Configuration Evidence

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Department:
Computer Science and Engineering

Project Domain:
Salesforce CRM and Business Process Automation

Technology Used:
Salesforce Platform, Apex, Apex Triggers, Batch Apex, Scheduled Apex, Flows, Custom Objects, Reports and Dashboards

Institution:
Kongu Engineering College

Submitted in partial fulfillment of the academic requirements for the Computer Science and Engineering

User Story

HandsMen Threads is a Salesforce application for a mens fashion retailer. It centralizes customer records, products, inventory, sales orders, marketing campaigns, and customer communication so that teams work from one connected source of information.

As a sales user, I want to manage customers and orders so that I can provide a timely, personalized shopping experience. As an inventory user, I want to monitor stock quantities and receive low-stock notifications so that products remain available. As a marketing user, I want to identify loyalty tiers and run campaigns so that customers receive relevant offers.

1 Introduction

1.1 Project Overview

HandsMen Threads was configured as a Salesforce Lightning application. It includes custom objects for HandsMen Customer, HandsMen Product, HandsMen Order, Inventory, and Marketing Campaign. Custom tabs make each function accessible from the application navigation, while validation rules, flows, email templates, roles, profiles, and permission sets control the business process and access model.

1.2 Purpose

- Centralize customer, product, order, inventory, and campaign information.
- Improve order processing through validation and confirmation emails.
- Protect inventory accuracy through stock validation and low-stock alerts.
- Support customer retention using loyalty status automation and campaign records.
- Provide role-based access for sales, inventory, and marketing users.

2 Ideation Phase

2.1 Problem Statement

Mens fashion retail teams need current information about customers, stock, orders, and campaigns. When this data is fragmented, staff cannot confidently answer product availability questions, confirm purchases, or target loyal customers. HandsMen Threads addresses this with a single Salesforce workspace and automation for recurring work.

The customer problem statements for each user group are summarised below.

PS (User)	I am trying to	But	Because	Which makes me feel
PS-1 Sales user	Manage customers and orders efficiently	Order follow-up is manual and customer history is scattered	Customer, order, and communication data are not connected	Frustrated and error-prone
PS-2 Inventory user	Keep stock figures accurate	Low stock is noticed too late	Stock levels are not monitored automatically	Worried and reactive
PS-3 Marketing user	Target loyal customers with campaigns	Campaigns lack customer context	Loyalty tiers are not updated systematically	Uncertain about results
PS-4 Customer	Know that my order is confirmed	I cannot tell whether my order was accepted	No automatic confirmation is sent	Anxious and uninformed

2.2 Empathy Map

User	Think and feel	Pain point	Salesforce response
Sales user	Needs a complete customer view	Order follow-up is manual	Customer and order records with confirmation email
Inventory user	Needs reliable stock figures	Low stock may be missed	Inventory object, validation, and alert flow

Marketing user	Needs audience segments	Campaigns lack customer context	Loyalty status and campaign relationship
Customer	Needs fast confirmation	Order status is unclear	Order confirmation communication

2.3 Brainstorming and Idea Prioritization

Step 1: Brainstormed ideas

- **Spreadsheet tracking:** Use shared spreadsheets to record customers, orders, and stock.
- **Standalone e-commerce website:** Build a custom web store for browsing and ordering.
- **POS or ERP extension:** Extend an existing point-of-sale or ERP system with retail modules.
- **Salesforce-based retail application:** Build one cloud application using custom objects, flows, validation rules, email templates, and roles.
- **Automated email notifications:** Send order confirmations and low-stock alerts automatically.
- **Loyalty segmentation automation:** Update customer loyalty status on a schedule from total purchases.

Step 2: Grouping

Group	Ideas	Description
Manual solutions	Spreadsheet tracking	Low cost and quick to start, but depends on manual updates and is prone to errors
Semi-automated solutions	Standalone e-commerce website, POS or ERP extension	Automate parts of the process, but need separate systems, heavy customization, or integration
Fully automated cloud solutions	Salesforce-based retail application, automated email notifications, loyalty segmentation automation	End-to-end automation of orders, stock alerts, and loyalty on one scalable platform

Step 3: Idea prioritization

Each idea is rated High, Medium, or Low against the criteria below.

Idea	Impact	Feasibility	Cost efficiency
Spreadsheet tracking	Low	High	High
Standalone e-commerce website	Medium	Medium	Low
POS or ERP extension	Medium	Low	Low
Salesforce-based retail application	High	High	High
Automated email notifications	High	High	High
Loyalty segmentation automation	High	Medium	High

3 Requirement Analysis

3.1 Functional Requirements

ID	Requirement	Implementation evidence
FR 1	Manage customer profiles and loyalty details	HandsMen Customer custom object with email, phone, loyalty status, total purchases, and name fields
FR 2	Manage products and stock	HandsMen Product and Inventory objects with SKU, price, stock quantity, warehouse, and stock status
FR 3	Record and validate orders	HandsMen Order object with customer lookup, quantity, status, total amount, and validation rule

FR 4	Communicate order status	Record-triggered order confirmation flow and email template
FR 5	Notify inventory users	Inventory record-triggered low-stock alert flow
FR 6	Segment loyalty customers	Scheduled loyalty status update flow using a loop and decision paths
FR 7	Restrict access	Sales, Inventory, and Marketing roles with profile and permission-set configuration

3.2 Non Functional Requirements

- Security: Users are assigned roles and platform profiles; permission sets can grant additional access.
- Data quality: Validation rules prevent invalid totals and stock values.
- Usability: Tabs group customer, order, inventory, product, reporting, and dashboard operations in one app.
- Maintainability: Salesforce flows use separate named automation for order confirmation, low stock, and loyalty status.

3.3 Customer Journey Map

The journey map follows a customer and the HandsMen Threads staff from first contact to loyalty engagement.

Stage 1: Discovery and Registration

- **User actions:** Learns about the brand and shares contact details in store or online.
- **Touchpoints:** Store, website, sales user.
- **Pain points:** Duplicate or incomplete customer details.
- **HandsMen Threads solution:** A HandsMen Customer record stores email, phone, name, loyalty status, and total purchases, with email validation.
- **Emotion:** Curious and hopeful.

Stage 2: Product Selection

- **User actions:** Browses the collection and checks price and availability.
- **Touchpoints:** Store, catalogue, sales user.
- **Pain points:** Stock availability is unclear.
- **HandsMen Threads solution:** HandsMen Product and Inventory records show SKU, price, stock quantity, and warehouse.
- **Emotion:** Interested but unsure.

Stage 3: Order Placement

- **User actions:** The sales user records the order with quantity and total amount.
- **Touchpoints:** Sales counter, sales user.
- **Pain points:** Manual entry mistakes and invalid totals.
- **HandsMen Threads solution:** A HandsMen Order record links to the customer, and the validation rule blocks a total of zero or less.
- **Emotion:** Hopeful.

Stage 4: Order Confirmation

- **User actions:** Waits to hear that the order is accepted.
- **Touchpoints:** Email.
- **Pain points:** Uncertainty about whether the order was received.
- **HandsMen Threads solution:** The Order Confirmation Flow sends an email alert using an email template.
- **Emotion:** Reassured.

Stage 5: Stock Monitoring

- **User actions:** The inventory user updates stock after each sale.

- **Touchpoints:** Inventory tab, email alerts.
- **Pain points:** Low stock is noticed late and invalid stock values are entered.
- **HandsMen Threads solution:** The inventory validation rule and the Low-Stock Alert Flow protect stock accuracy.
- **Emotion:** In control.

Stage 6: Loyalty and Campaigns

- **User actions:** The marketing user reviews loyalty tiers and plans campaigns.
- **Touchpoints:** Customer records, Marketing Campaign records, loyalty email.
- **Pain points:** No way to segment loyal customers.
- **HandsMen Threads solution:** The Scheduled Loyalty Status Update Flow sets loyalty status, and campaigns link to customers.
- **Emotion:** Satisfied and confident.

3.4 Data Flow Diagram

This is the Level-1 data flow diagram for the main steps of the system.

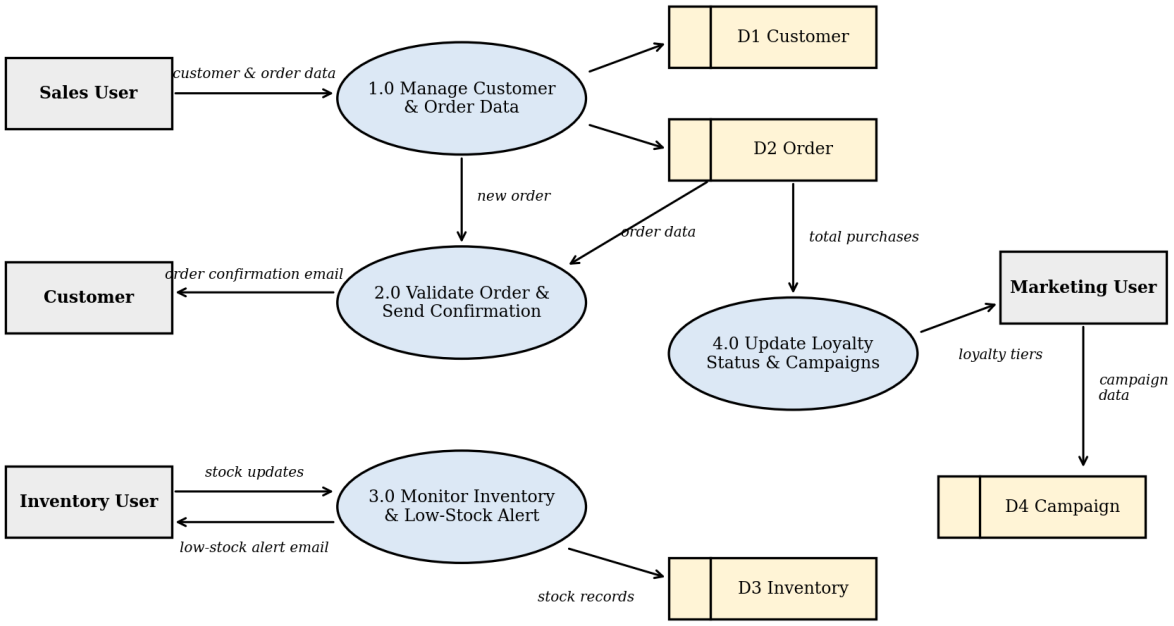


Figure: Level-1 data flow diagram for HandsMen Threads.

Here is how the diagram maps to the steps:

- **Step 1, input:** The sales user submits customer and order data into process 1.0, which stores it in D1 (Customer) and D2 (Order).
- **Step 2, validation and confirmation:** Process 2.0 reads the order data, applies the validation rule, and sends a confirmation email to the customer.
- **Step 3, inventory control:** Process 3.0 receives stock updates from the inventory user, saves them in D3 (Inventory), and returns a low-stock alert email.
- **Step 4, loyalty and campaigns:** Process 4.0 reads total purchases, updates loyalty status, and supports marketing campaign records in D4 (Campaign).

4 Project Design

4.1 Solution Architecture

Layer	HandsMen Threads design
Presentation	Salesforce Lightning application, object tabs, lists, reports, and dashboards
Business automation	Order confirmation flow, low-stock alert flow, scheduled loyalty update flow
Data	HandsMen Customer, HandsMen Product, HandsMen Order, Inventory, Marketing Campaign
Security	Platform profile, permission set, and Sales, Inventory, and Marketing roles
Communication	Classic email templates for order confirmation and loyalty engagement

4.2 Data Model

Object	Purpose	Key fields and relationships
HandsMen Customer	Stores customer and loyalty information	Email, phone, first name, last name, full name, loyalty status, total purchases
HandsMen Product	Stores retail product information	Price, SKU, stock quantity, and order lookup
HandsMen Order	Records a customer purchase	Customer lookup, quantity, status, total amount
Inventory	Tracks stock by warehouse	Product master-detail, stock quantity, stock status, warehouse
Marketing Campaign	Tracks customer-focused promotions	Customer lookup, start date, end date

4.3 Controls and Automation

The order validation rule blocks a total amount less than or equal to zero. The inventory validation rule blocks a stock quantity less than or equal to zero. Customer email validation is configured to enforce the project email condition. The order and inventory flows send email alerts; evidence records completed debug runs for both automations. The scheduled loyalty flow segments records using total-purchase decision paths.

4.4 Problem-Solution Fit

HandsMen Threads addresses fragmented retail processes by matching user needs and constraints to Salesforce-native capabilities.

Problem (user behavior or constraint)	HandsMen Threads solution	Salesforce-native capability
Customer, order, and stock data are scattered	Centralized customer, product, order, inventory, and campaign records	Custom objects with lookup and master-detail relationships
Invalid totals and stock values are entered	Data is checked at the point of entry	Validation rules
Order confirmation is manual	Confirmation email is sent automatically when an order is saved	Record-triggered flow and email template
Low stock is noticed late	Inventory users are alerted automatically	Record-triggered flow and email alert
Loyal customers cannot be identified	Loyalty status is updated from total purchases	Scheduled flow with loop and decision elements
Staff can see data they do not need	Users see only what their role permits	Roles, profile, and permission set
Different teams need different views	One app with a tab for each business function	Lightning app and custom tabs

4.5 Proposed Solution

Parameter	Description
Problem	Fragmented customer, order, stock, and campaign handling leads to manual follow-up, late stock alerts, and untargeted marketing.
Solution	A Salesforce Lightning application that manages customers, products, orders, inventory, and campaigns with validation, email automation, and role-based access.
Novelty	End-to-end automation of order confirmation, low-stock alerts, and loyalty segmentation using Salesforce-native tools.
Business impact	Faster order confirmation, fewer stock-outs, and loyalty-based targeting that supports customer retention.
Revenue model	Implementation, licensing, and support for fashion and retail businesses.

Parameter	Description
Scalability	New products, warehouses, users, and campaigns can be added through configuration without redesign.

5 Project Planning and Scheduling

5.1 Project Planning

The project follows the **Agile methodology**. Work is organized as Epics, User Stories, and Story Points and delivered in sprints. Velocity from completed sprints is used to estimate the remaining work.

Product Backlog and Sprint Schedule

Sprint	Functional requirement (epic)	User story number	User story / task	Story points	Priority	Team member
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to create a Salesforce Developer Edition org so that I can configure and test HandsMen Threads.	1	High	Deshaanth M.
Sprint-1	Data Modeling	USN-2	As an admin, I want custom objects for HandsMen Customer, HandsMen Product, HandsMen Order, Inventory, and Marketing Campaign so that retail data is stored in one place.	3	High	Deshaanth M.
Sprint-2	Data Modeling	USN-3	As a user, I want a dedicated tab for each object so that I can move easily between customers, orders, products, inventory, and campaigns.	3	High	Deshaanth M.
Sprint-2	Data Modeling	USN-4	As an admin, I want fields such as email, phone, loyalty status, SKU, price, stock quantity, and order status so that complete records are captured.	5	High	Deshaanth M.
Sprint-2	Data Modeling	USN-5	As a sales user, I want a Lightning app so that I can reach all retail operations from one navigation bar.	3	High	Deshaanth M.
Sprint-2	Data Modeling	USN-6	As an admin, I want lookup and master-detail relationships between customers, orders, products, inventory, and campaigns so that related records are linked.	5	Medium	Deshaanth M.
Sprint-3	Validation and Automation	USN-7	As a system, I want validation rules on order totals, stock quantities, and customer email so that invalid records are blocked.	3	High	Deshaanth M.
Sprint-3	Validation and Automation	USN-8	As a customer, I want an automatic order confirmation email so that I know my order was received.	5	High	Deshaanth M.
Sprint-3	Validation and Automation	USN-9	As an inventory user, I want a low-stock alert flow so that I am notified before products run out.	3	High	Deshaanth M.
Sprint-3	Validation and Automation	USN-10	As a marketing user, I want a scheduled flow that updates loyalty status from total purchases so that I can target loyal customers.	5	Medium	Deshaanth M.
Sprint-4	Security	USN-11	As an admin, I want Sales, Inventory, and Marketing roles with a platform profile and permission set so that users access only relevant data.	5	High	Deshaanth M.
Sprint-4	Apex and Testing	USN-12	As a developer, I want the InventoryBatchJob and OrderTriggerHandler Apex classes in the org so that custom logic can be extended.	3	Medium	Deshaanth M.
Sprint-4	Apex and Testing	USN-13	As a tester, I want debug runs of the flows and sample records so that the automation is verified.	3	High	Deshaanth M.

Epic Summary

Epic	Stories	Story points
Developer Setup	USN-1	1
Data Modeling	USN-2 to USN-6	19
Validation and Automation	USN-7 to USN-10	16
Security	USN-11	5
Apex and Testing	USN-12 to USN-13	6
Total	13 stories	47

6 Project Development Phase

This phase builds the solution milestone by milestone. Each milestone gives its definition and key steps, followed by the Salesforce screenshots that document the completed configuration.

Milestone 1: Salesforce Developer Account Creation

Definition: A free Salesforce Developer Edition org is the working environment where HandsMen Threads is built, tested, and demonstrated without affecting any production system.

Key steps:

- Sign up at developer.salesforce.com/signup
- Verify the email address and set a password
- Log in and open Setup

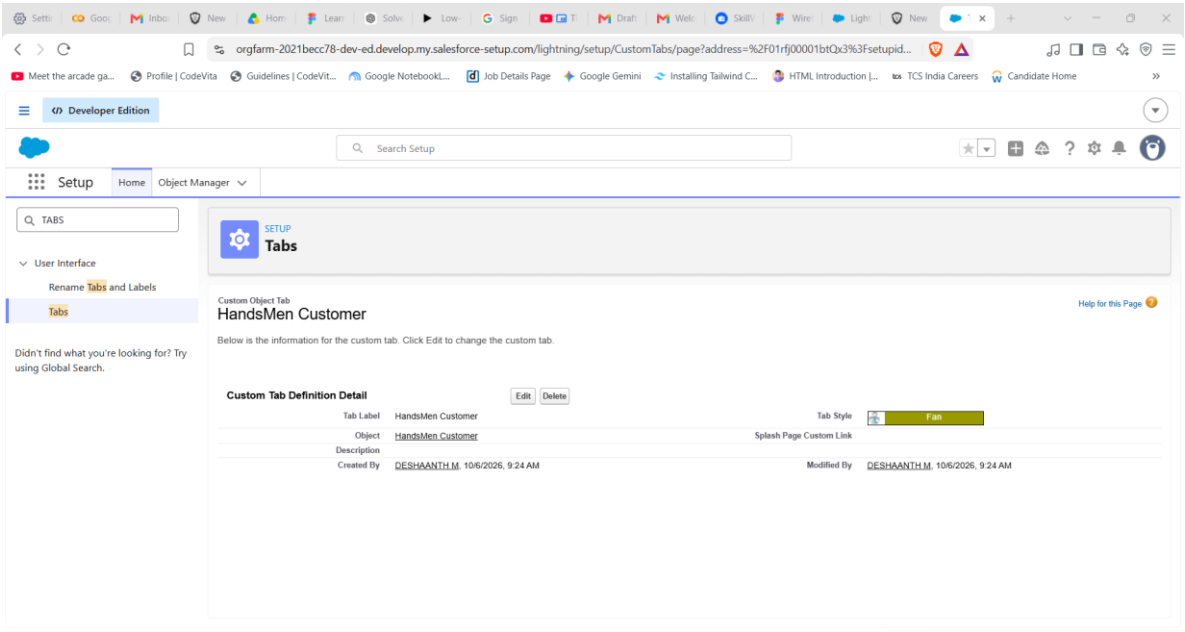


Figure: Custom tab for HandsMen Customer.

Milestone 2: Custom Object Creation

Definition: A custom object is a database table in Salesforce. Five objects store the retail data: HandsMen Customer, HandsMen Product, HandsMen Order, Inventory, and Marketing Campaign.

Key steps:

- Setup > Object Manager > Create > Custom Object
- Enter the label, plural label, and record name
- Allow Reports and Search, and launch the New Custom Tab wizard

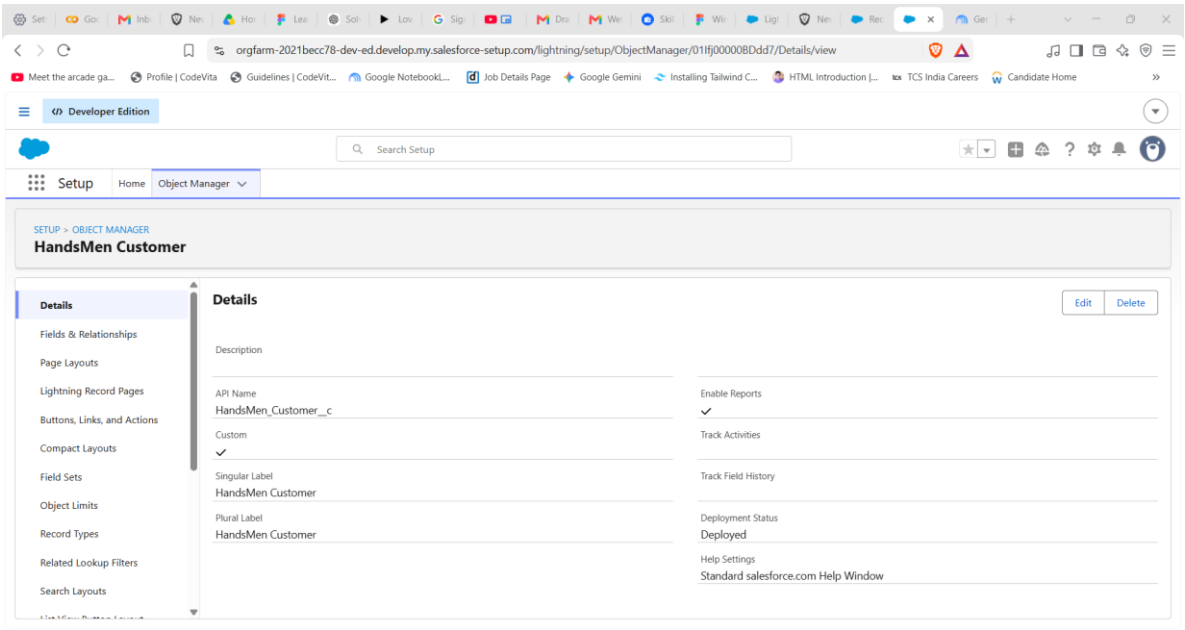


Figure: HandsMen Customer custom object details.

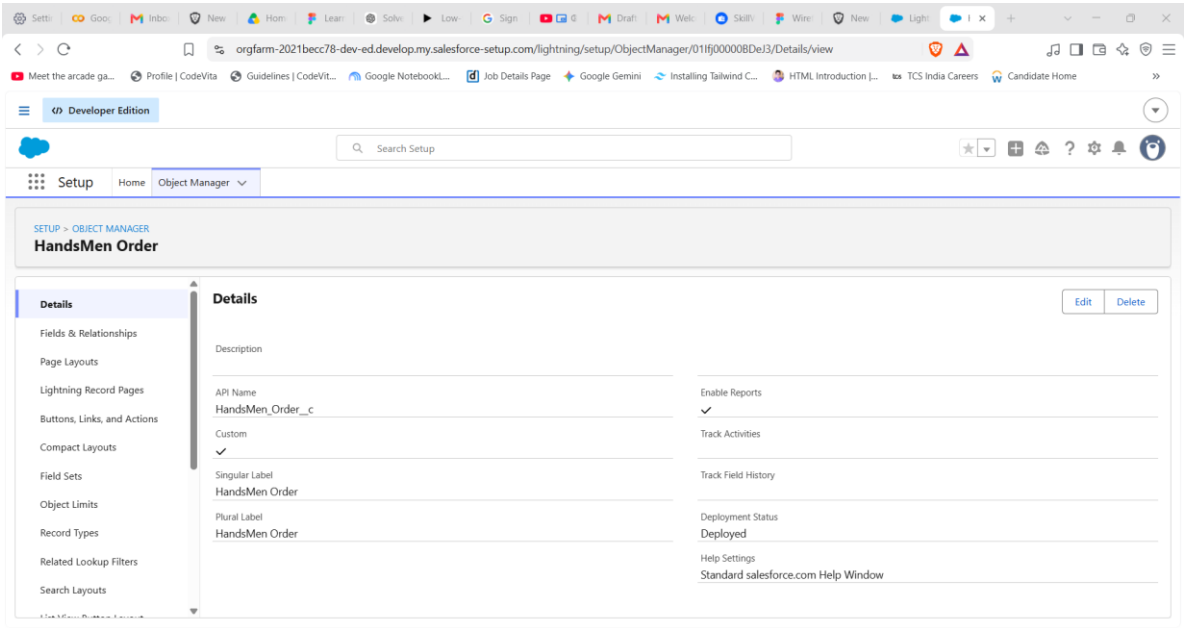


Figure: HandsMen Order object detail.

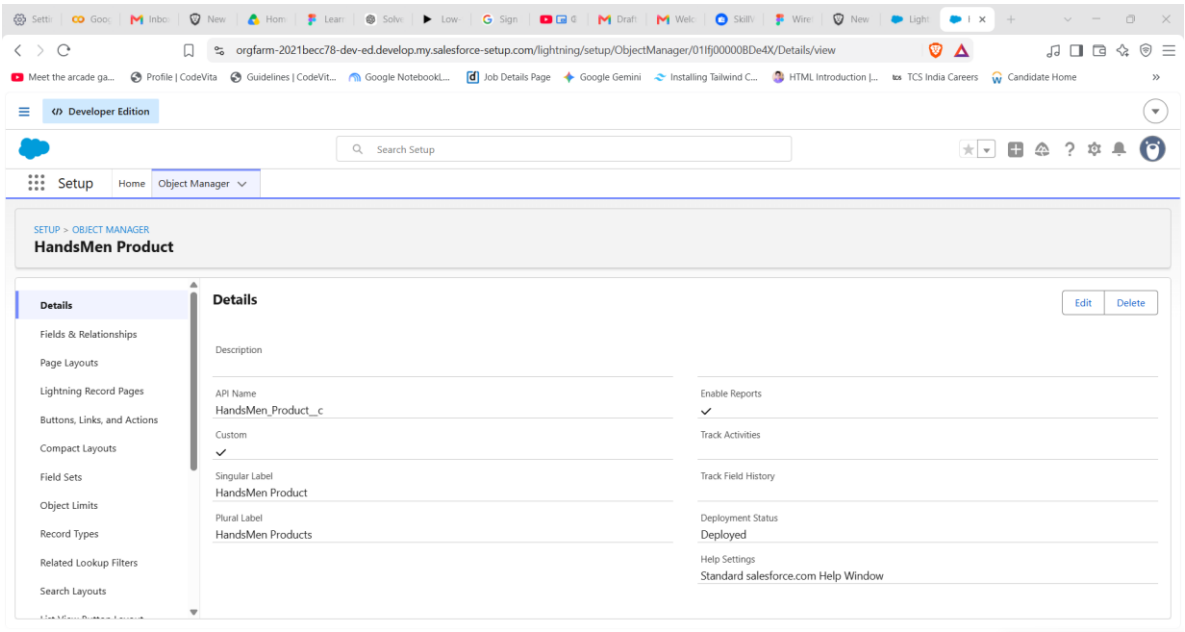


Figure: HandsMen Product object detail.

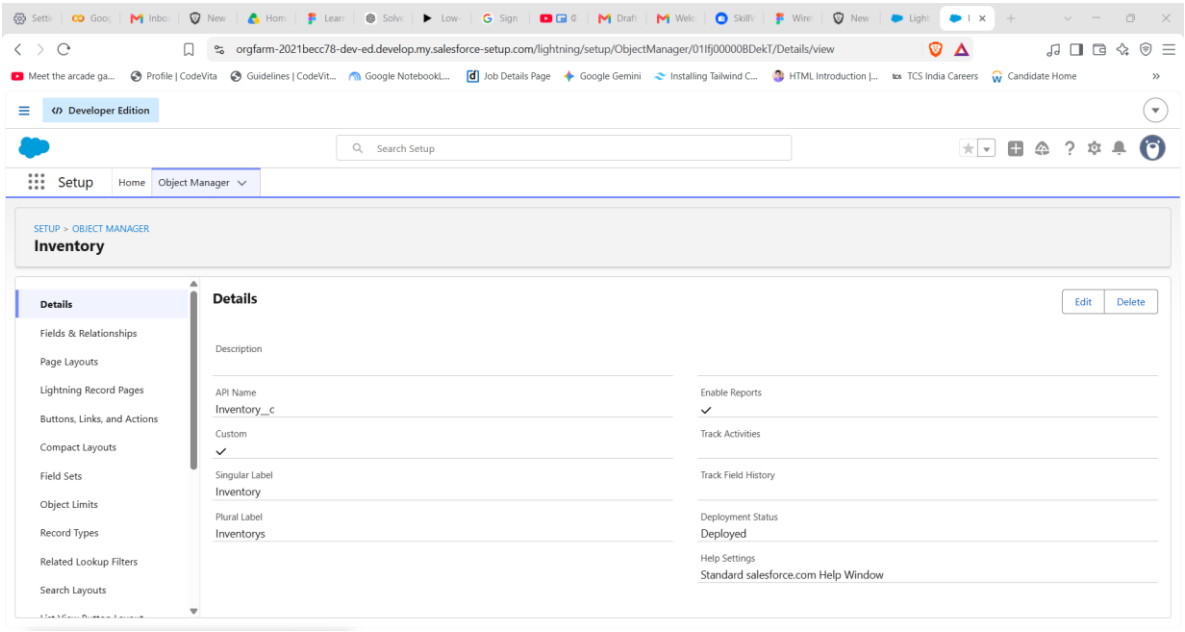


Figure: Inventory object detail.

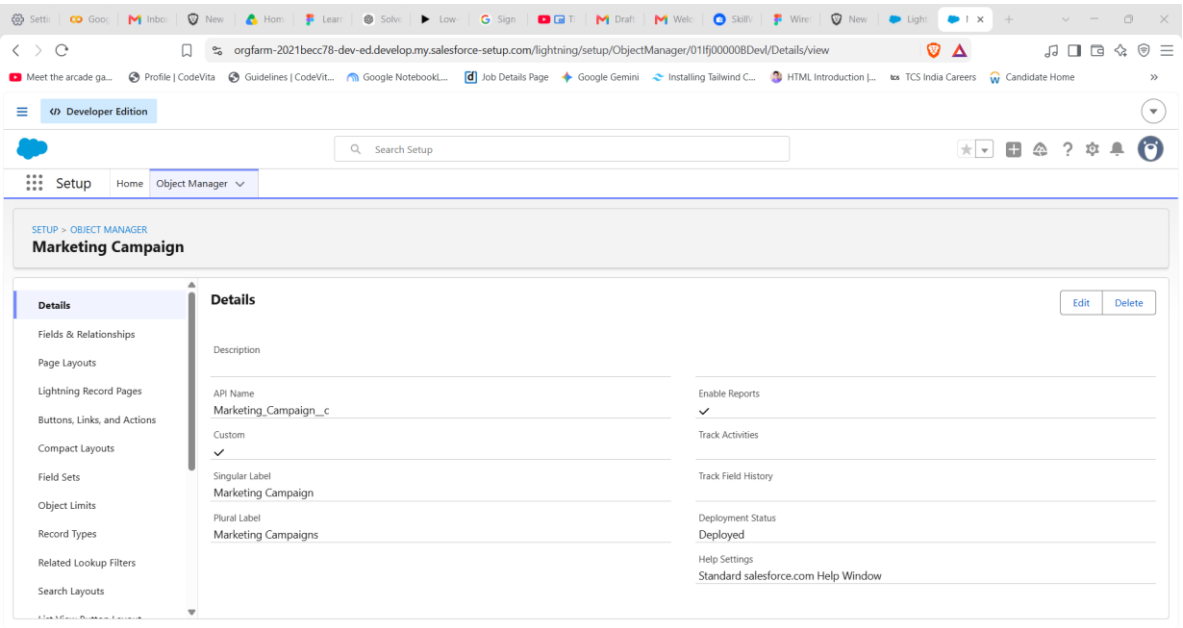


Figure: Marketing Campaign object detail.

Milestone 3: Tabs Creation

Definition: A tab is a navigation shortcut that opens an object's list of records. Dedicated tabs let users move quickly between customers, orders, products, inventory, and campaigns.

Key steps:

- Setup > Tabs > Custom Object Tabs > New
- Select the object and choose a tab style
- Set profile visibility and add the tab to the app

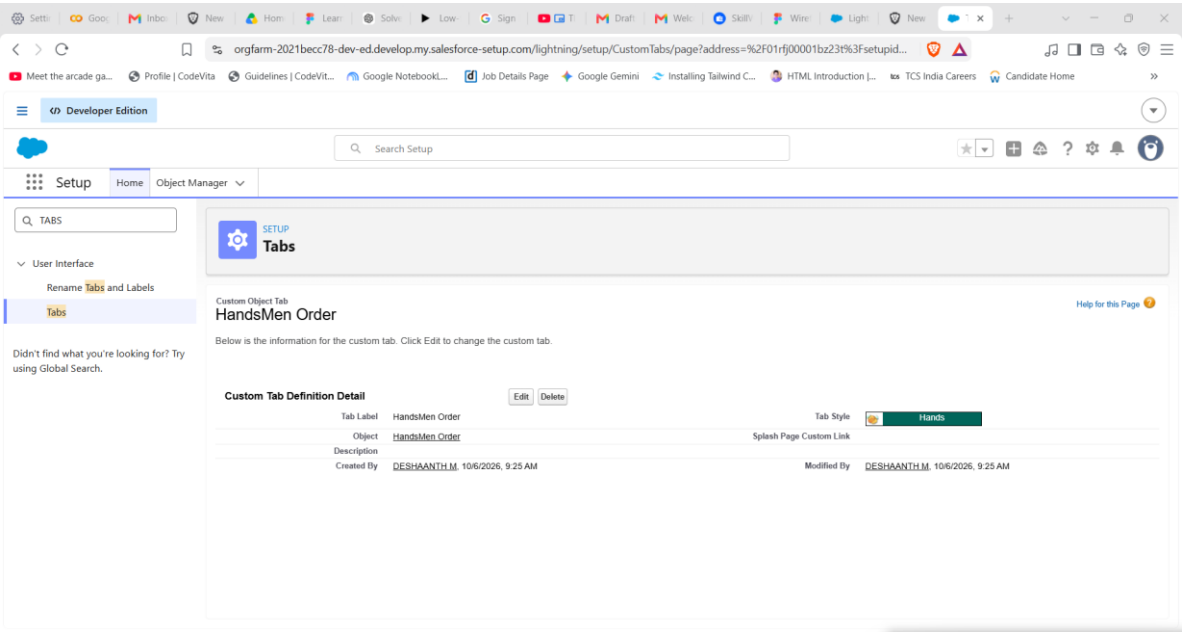


Figure: Custom tab for HandsMen Order.

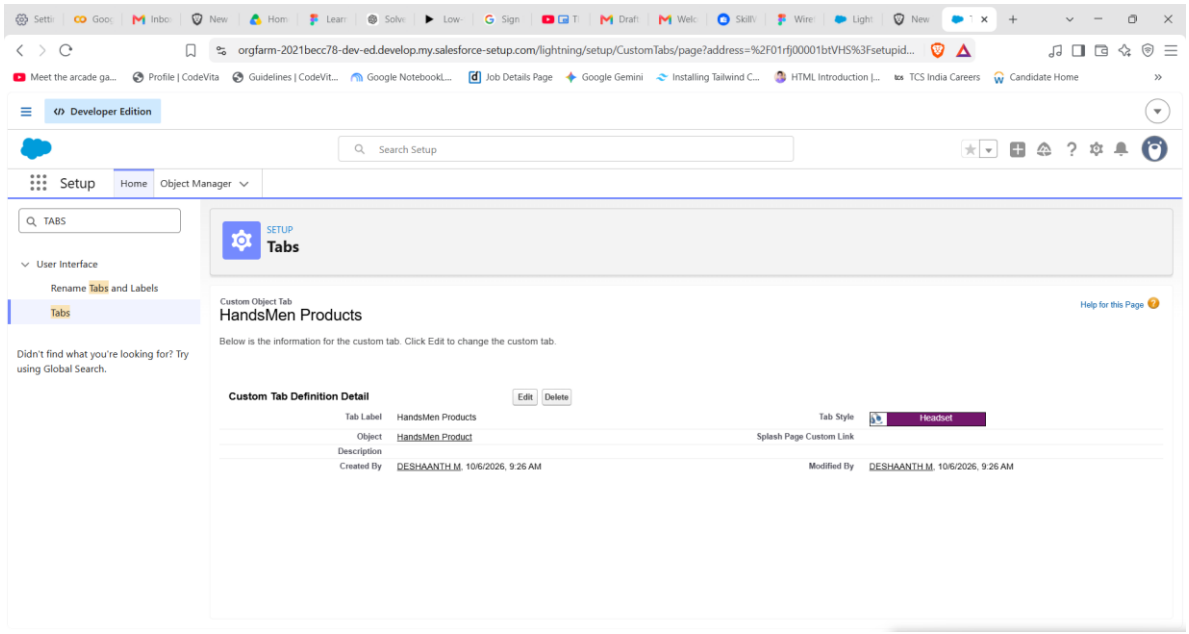


Figure: Custom tab for HandsMen Products.

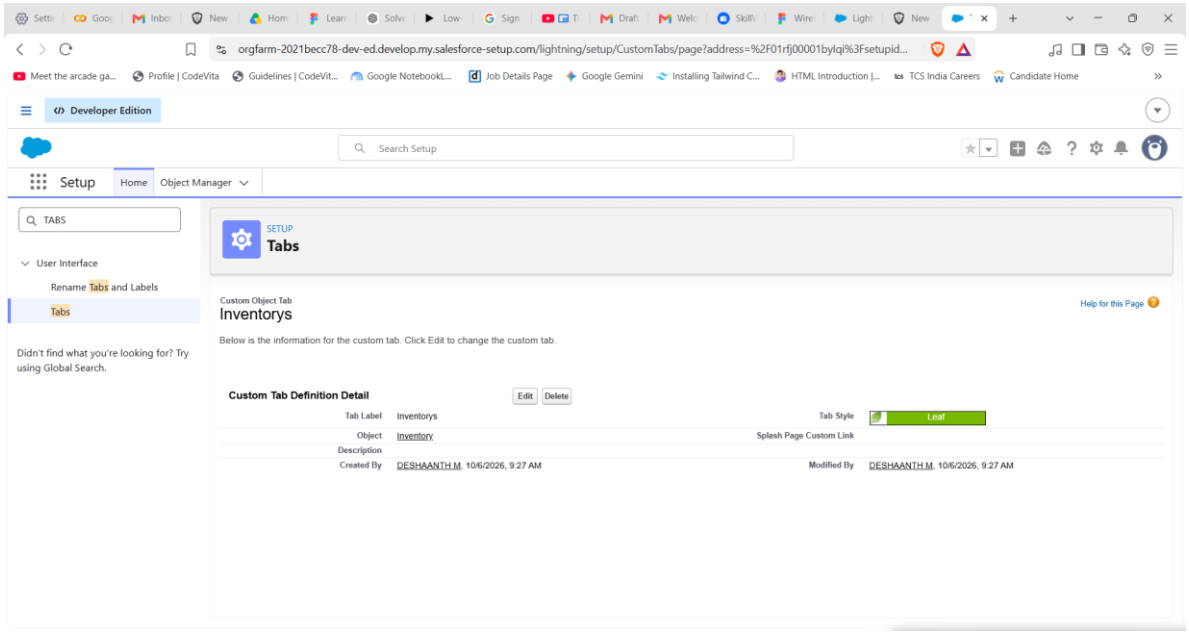


Figure: Custom tab for Inventory.

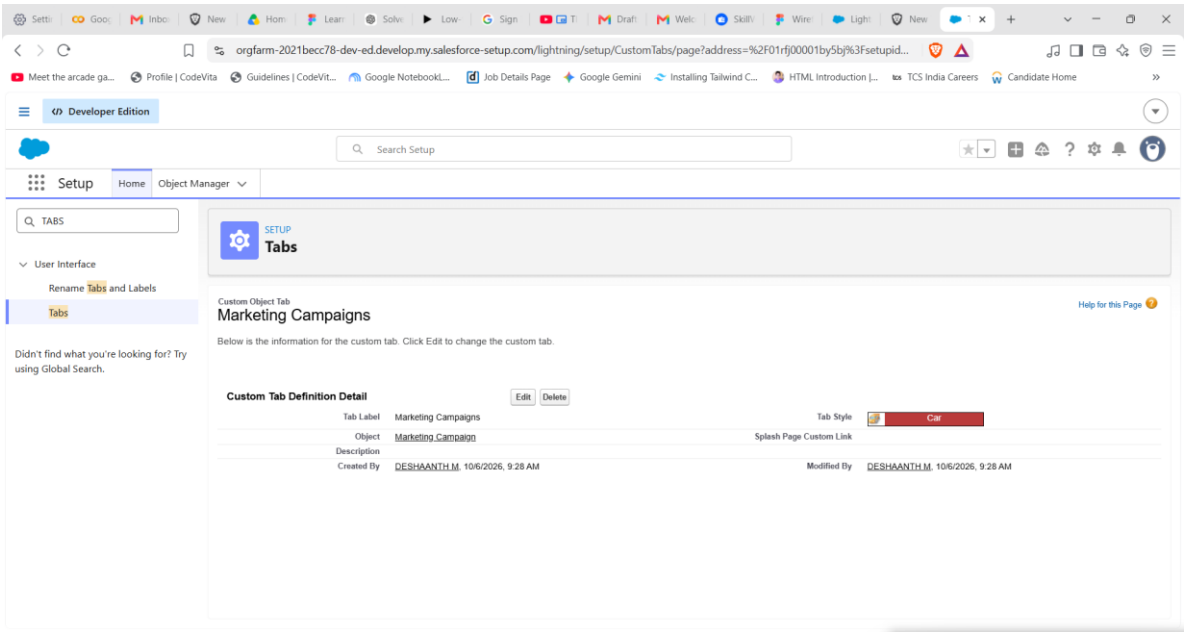


Figure: Custom tab for Marketing Campaigns.

Milestone 4: Lightning App Creation

Definition: A Lightning app groups tabs into one workspace. The HandsMen Threads app gives every team a single navigation bar.

Key steps:

- Setup > App Manager > New Lightning App
- Set the name and navigation style
- Add the object tabs to the app
- Assign the app to user profiles

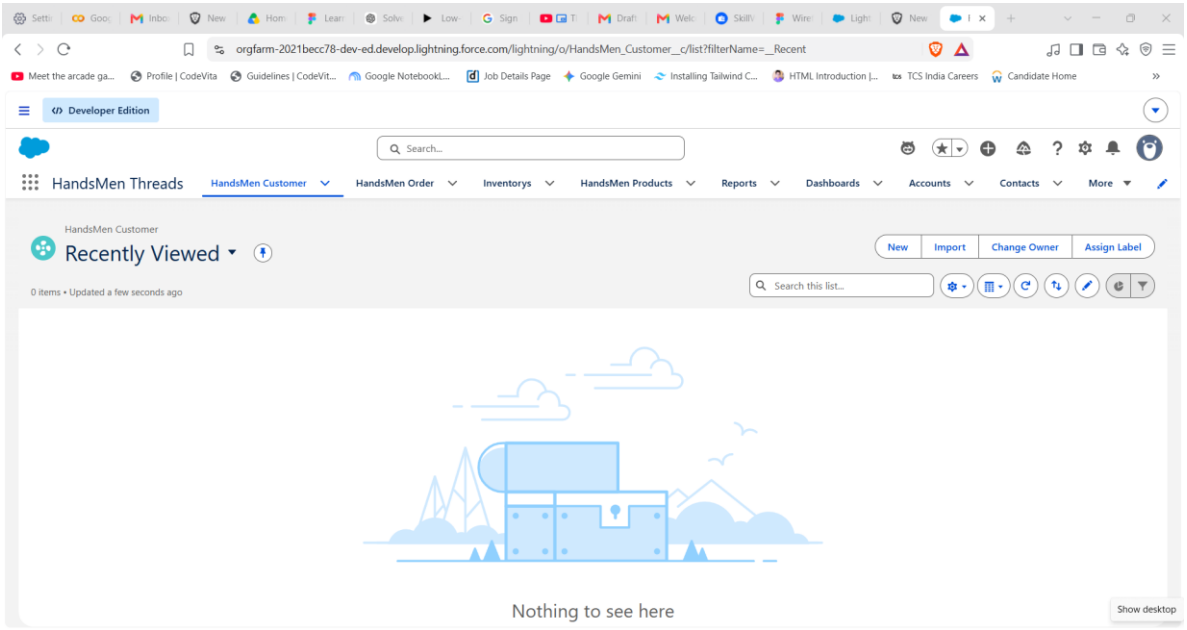


Figure: HandsMen Threads Lightning application navigation.

Milestone 5: Fields Creation

Definition: Fields are the columns of an object, and each holds one piece of data, such as email, phone, loyalty status, SKU, price, stock quantity, and order status.

Key steps:

- Object Manager > (object) > Fields & Relationships > New
- Choose the data type, such as Text, Email, Number, Currency, Picklist, or Formula
- Set the field name and required flag
- Set field-level security and add to layouts

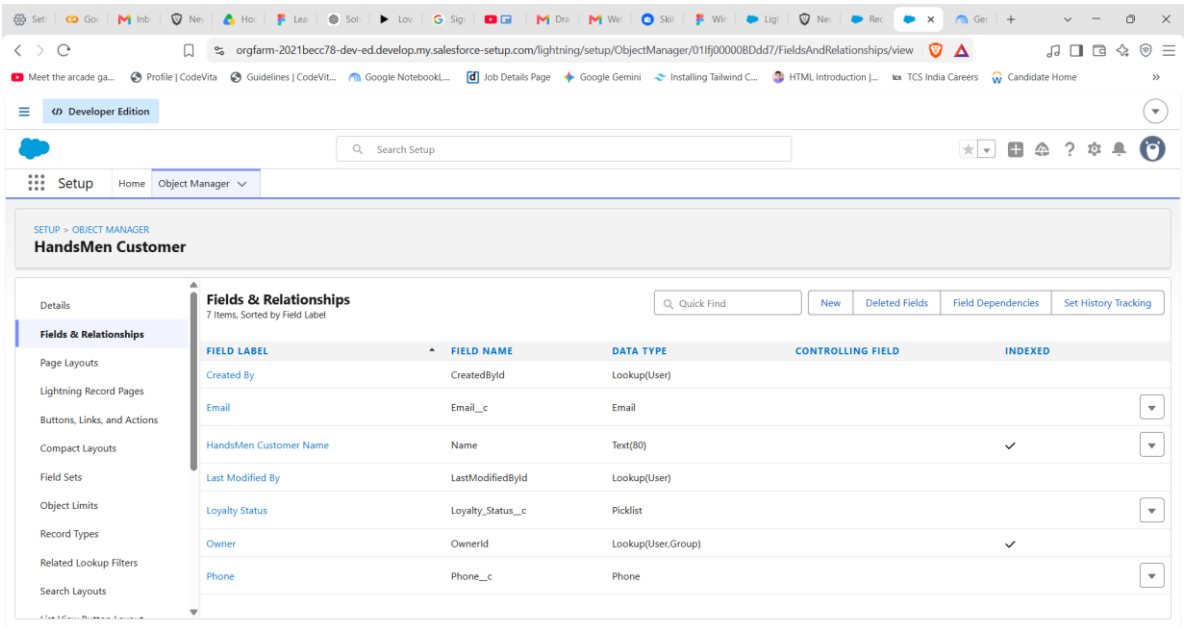


Figure: HandsMen Customer fields: email, loyalty status, and phone.

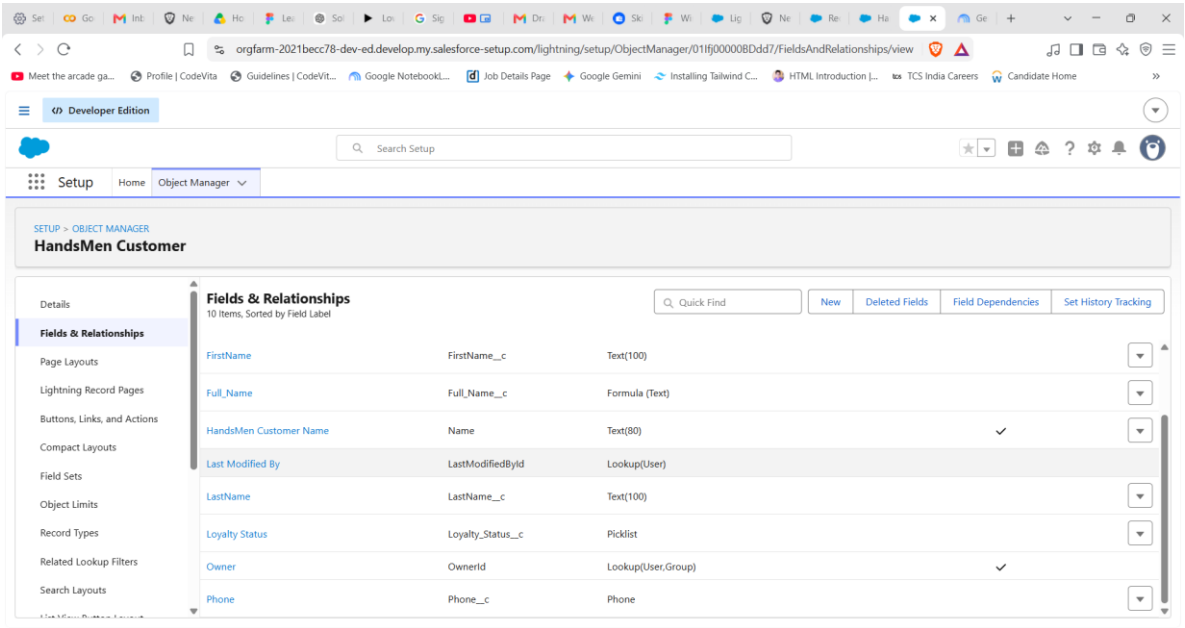


Figure: HandsMen Customer fields including first name, last name, and full-name formula.

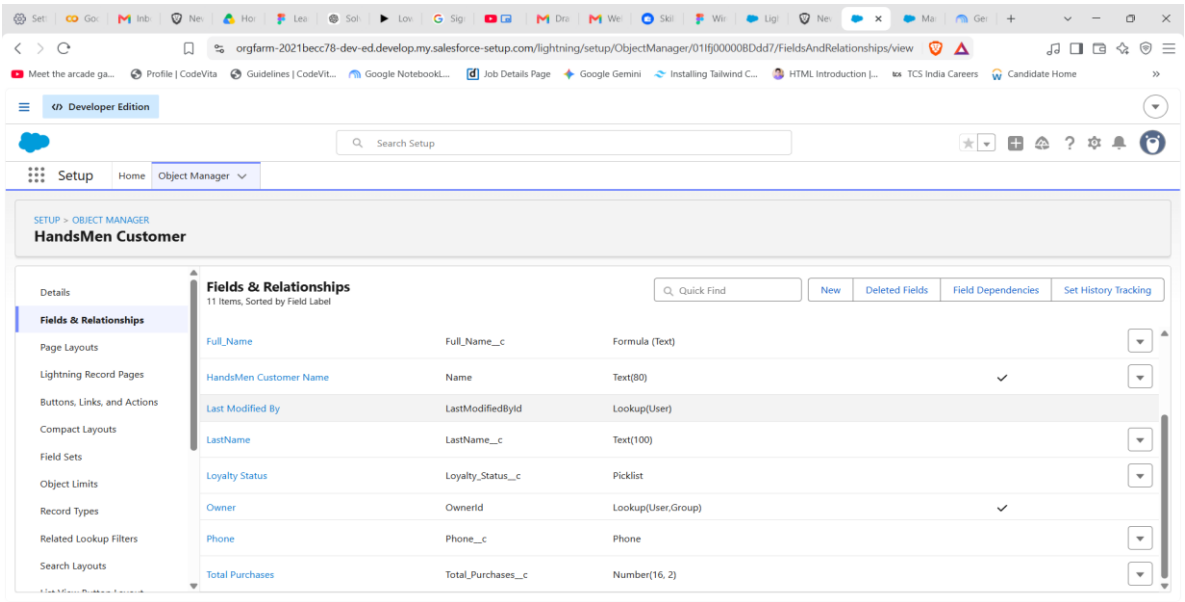


Figure: HandsMen Customer fields including total purchases.

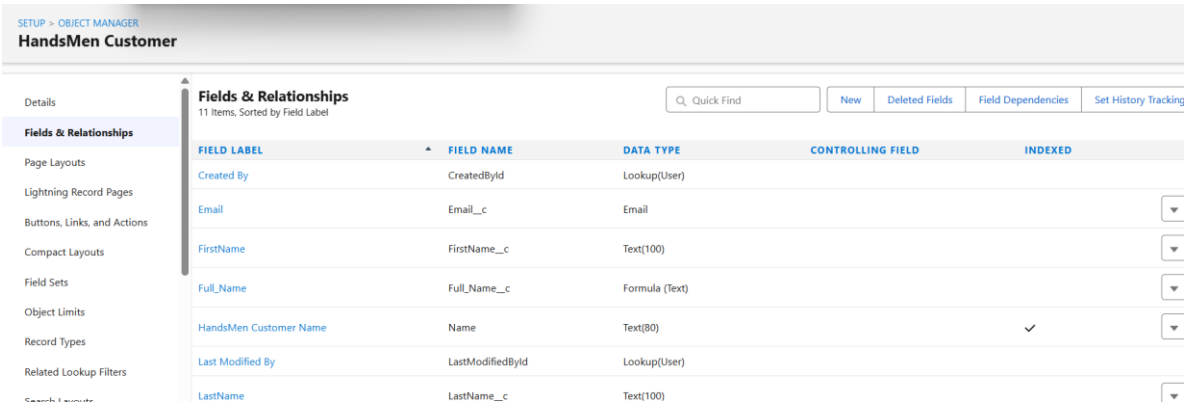


Figure: Expanded HandsMen Customer field list.

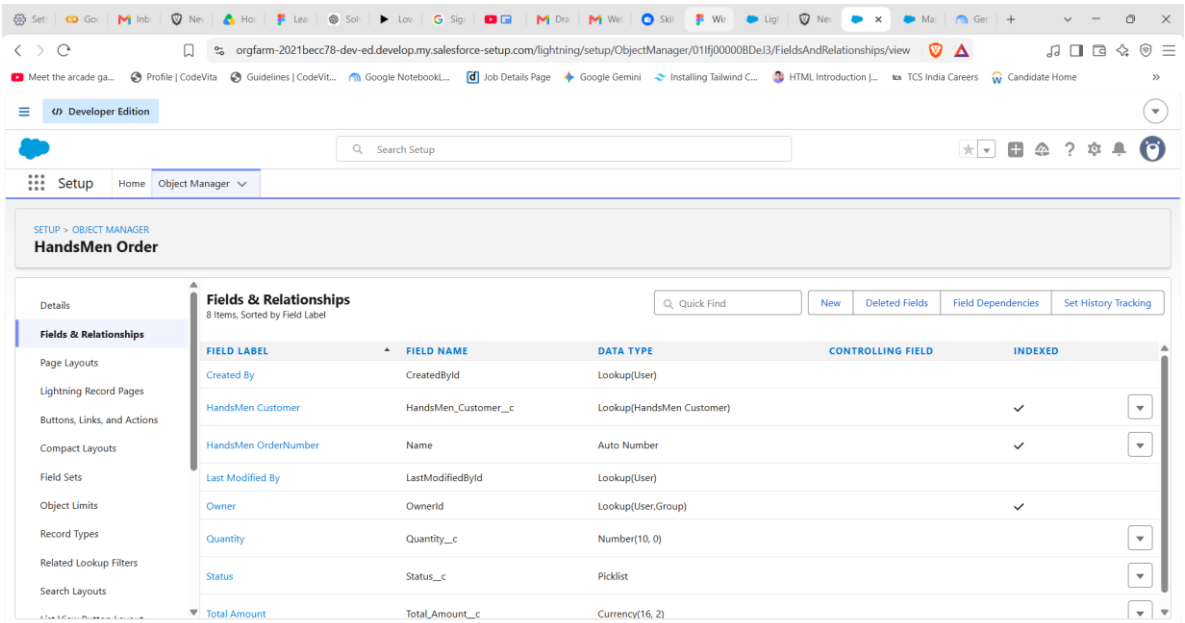


Figure: HandsMen Order fields: quantity, status, and total amount.

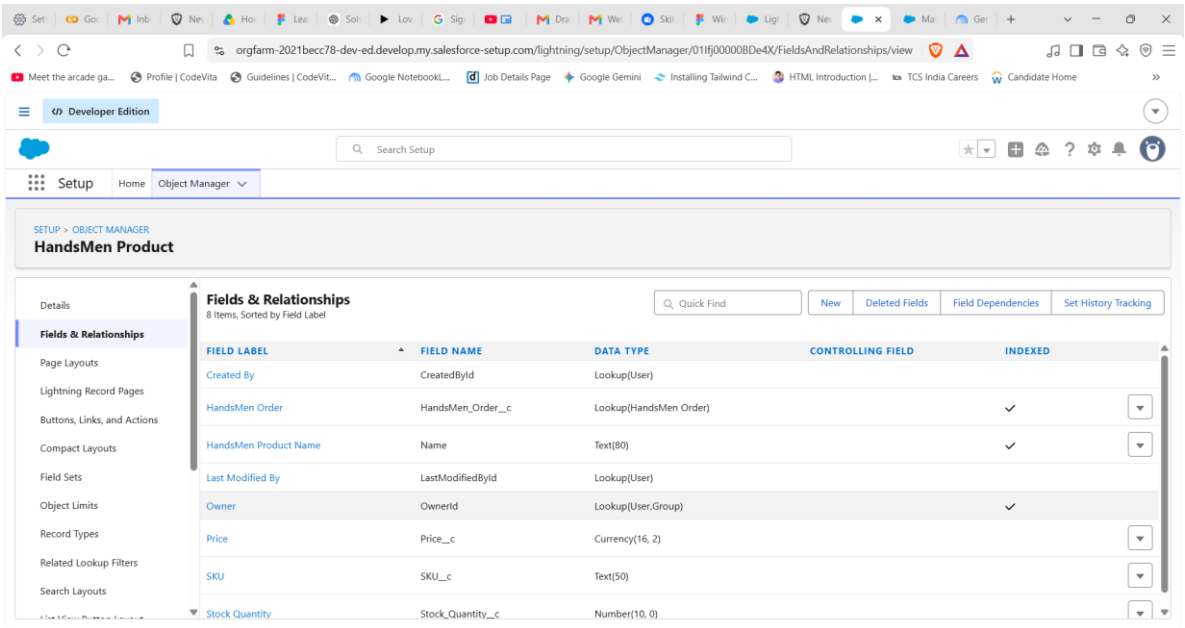


Figure: HandsMen Product fields: order relationship, price, SKU, and stock quantity.

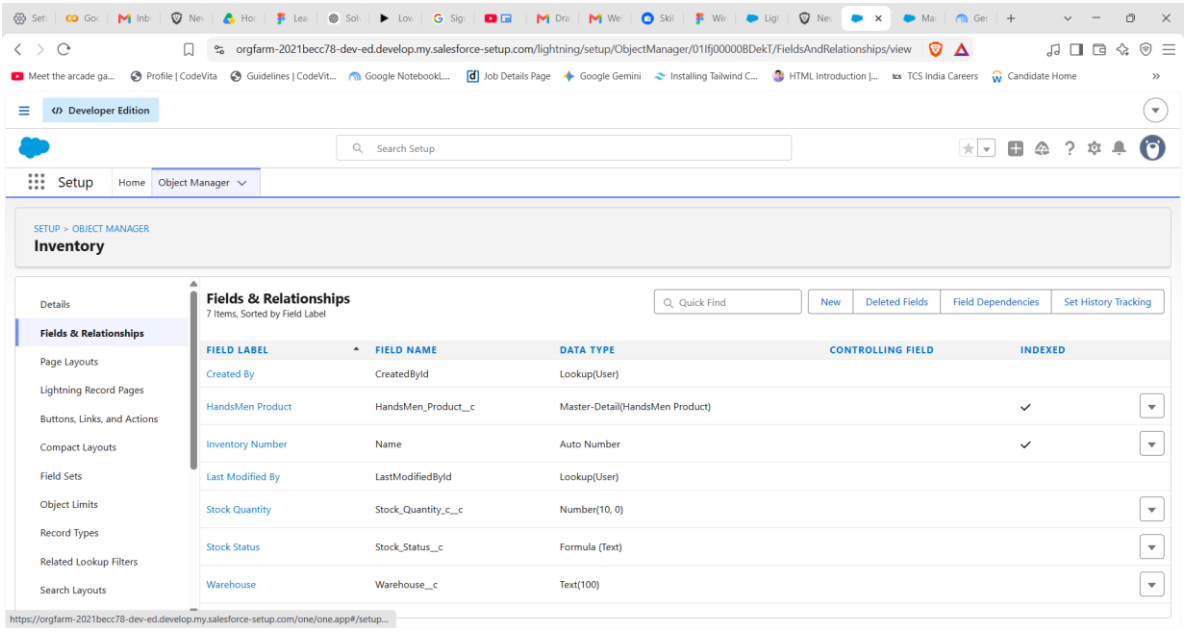


Figure: Inventory fields: product relationship, quantity, status, and warehouse.

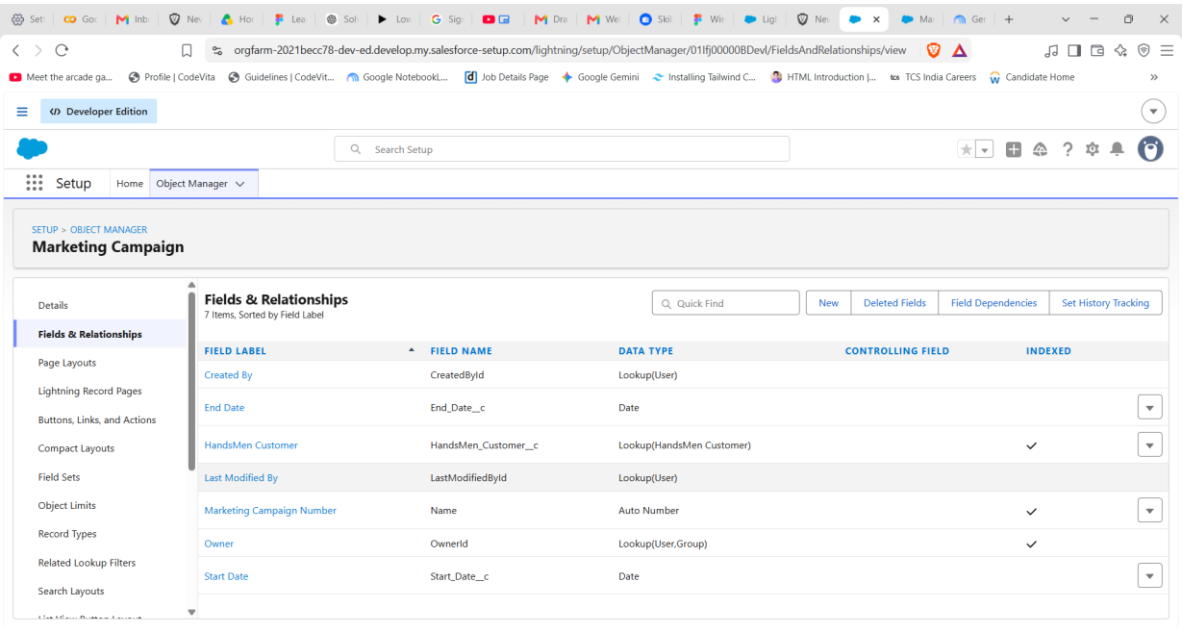


Figure: Marketing Campaign fields: customer relationship and campaign dates.

Milestone 6: Relationships

Definition: Relationships link records across objects so that activity can be tracked. Orders and campaigns look up a customer, and Inventory is a child of Product through a master-detail field.

Key steps:

- Create a lookup from HandsMen Order to HandsMen Customer
- Create a master-detail from Inventory to HandsMen Product
- Create a lookup from Marketing Campaign to HandsMen Customer
- Check the related lists on parent records

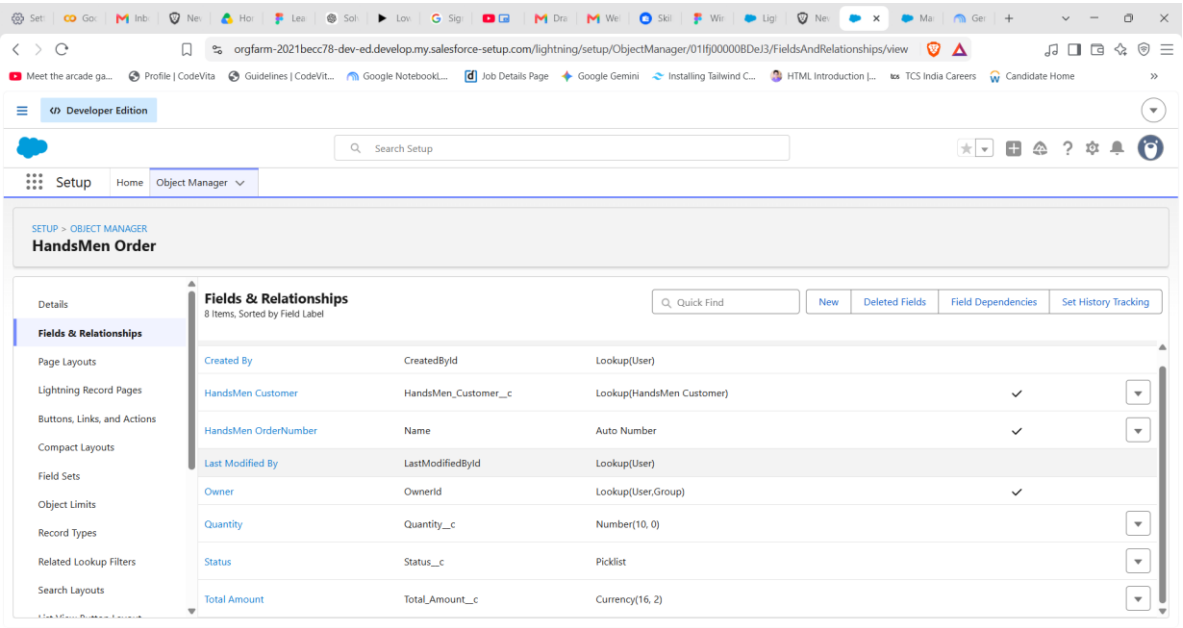


Figure: HandsMen Order fields and customer relationship.

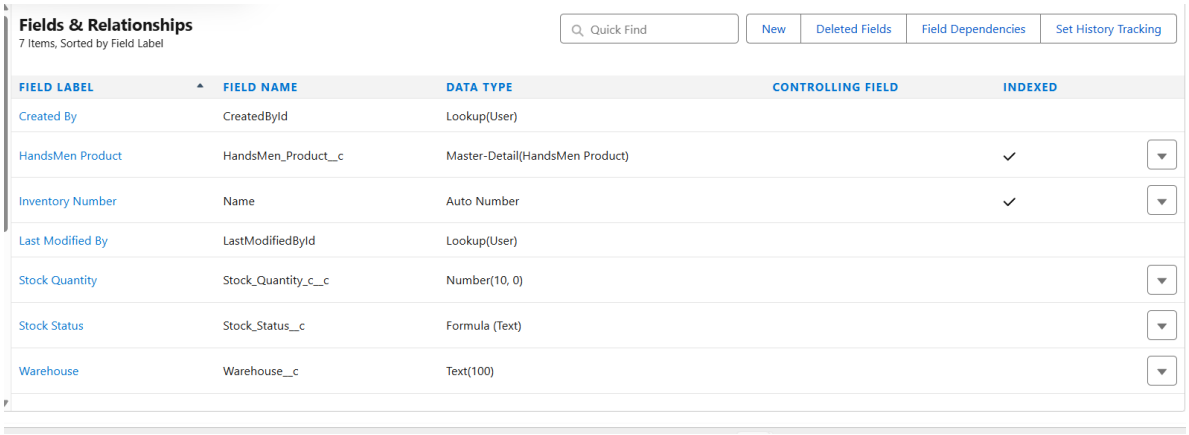


Figure: Inventory field configuration detail.

Milestone 7: Validation Rules

Definition: A validation rule is a formula that blocks a record from being saved when data is wrong. Rules cover order total, stock quantity, and customer email.

Key steps:

- Object Manager > (object) > Validation Rules > New
- Write the error condition formula, such as a total amount less than or equal to zero
- Enter the error message and its location
- Activate the rule and test with invalid data

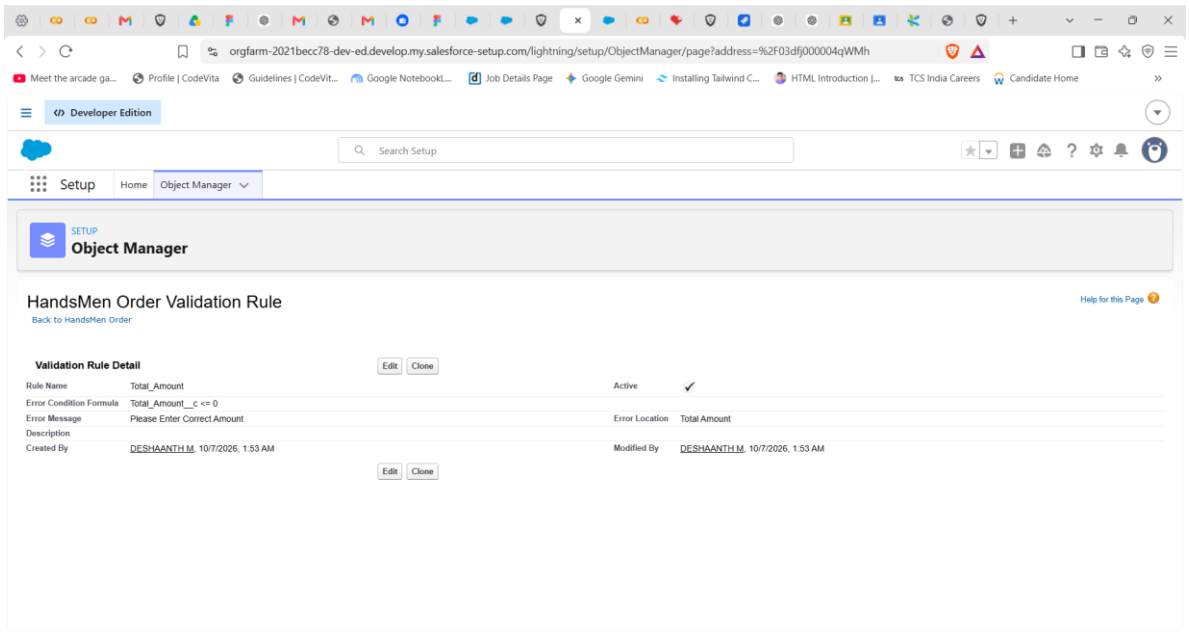


Figure: Order validation rule requiring a positive total amount.

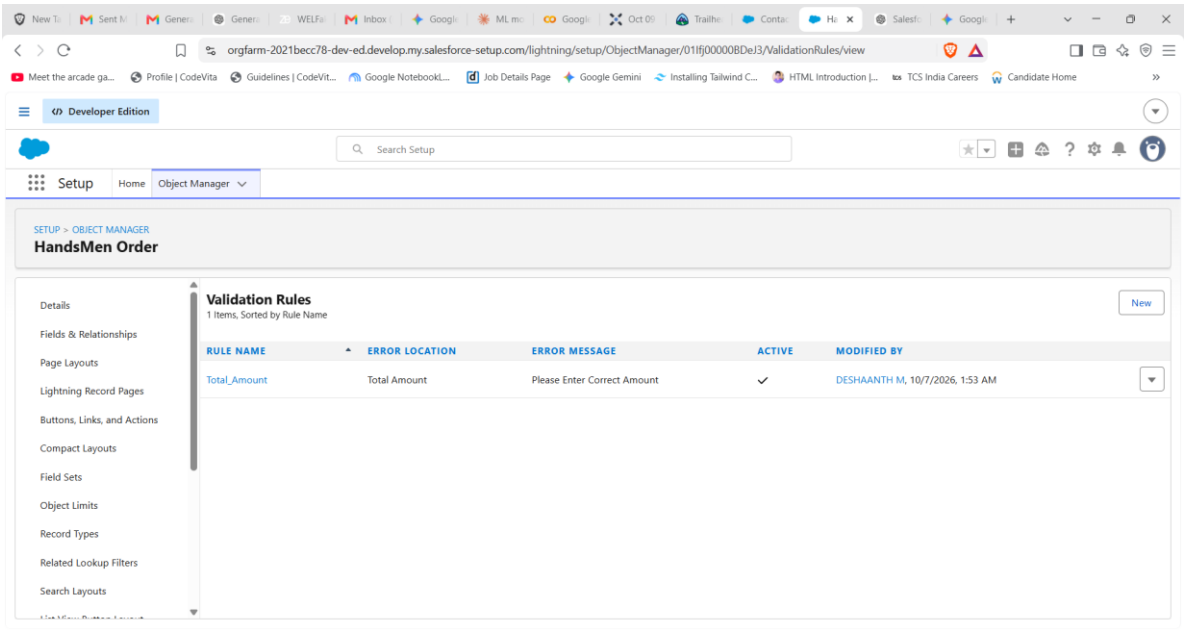


Figure: Active HandsMen Order validation-rule list for Total Amount.

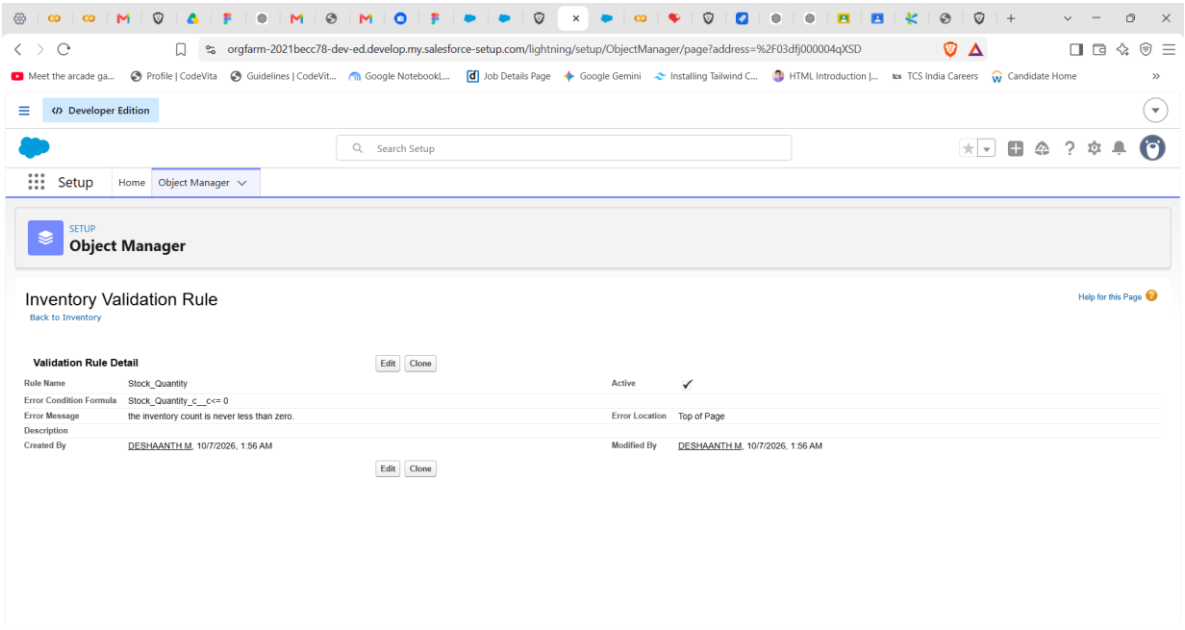


Figure: Inventory validation rule preventing nonpositive stock quantities.

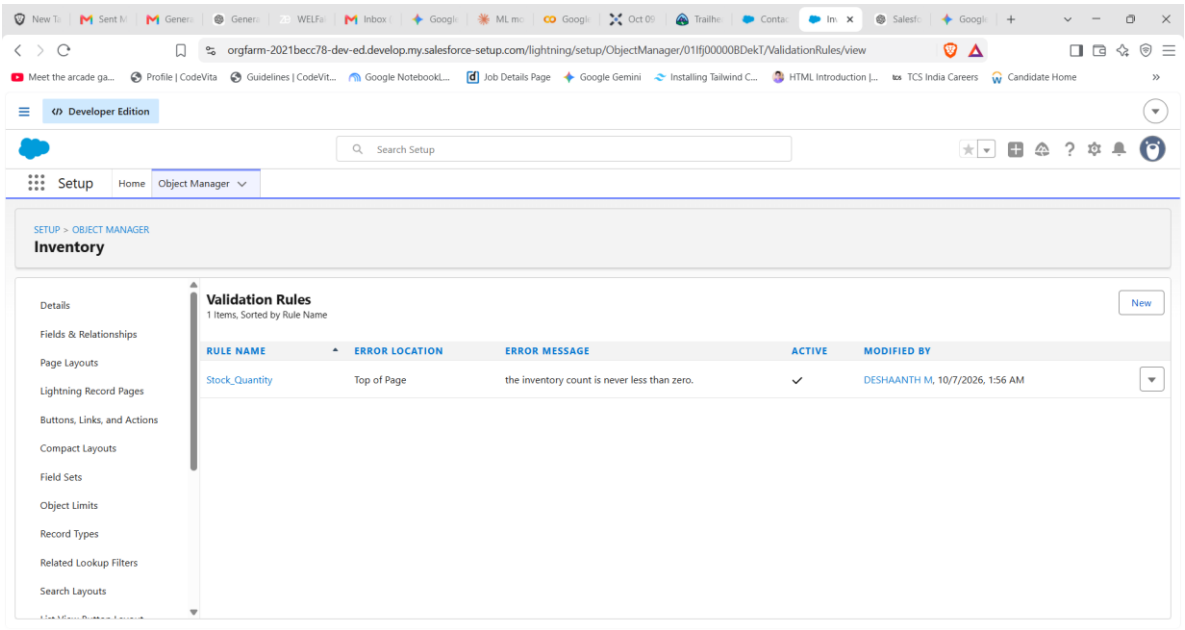


Figure: Active Inventory validation-rule list showing the Stock Quantity rule.

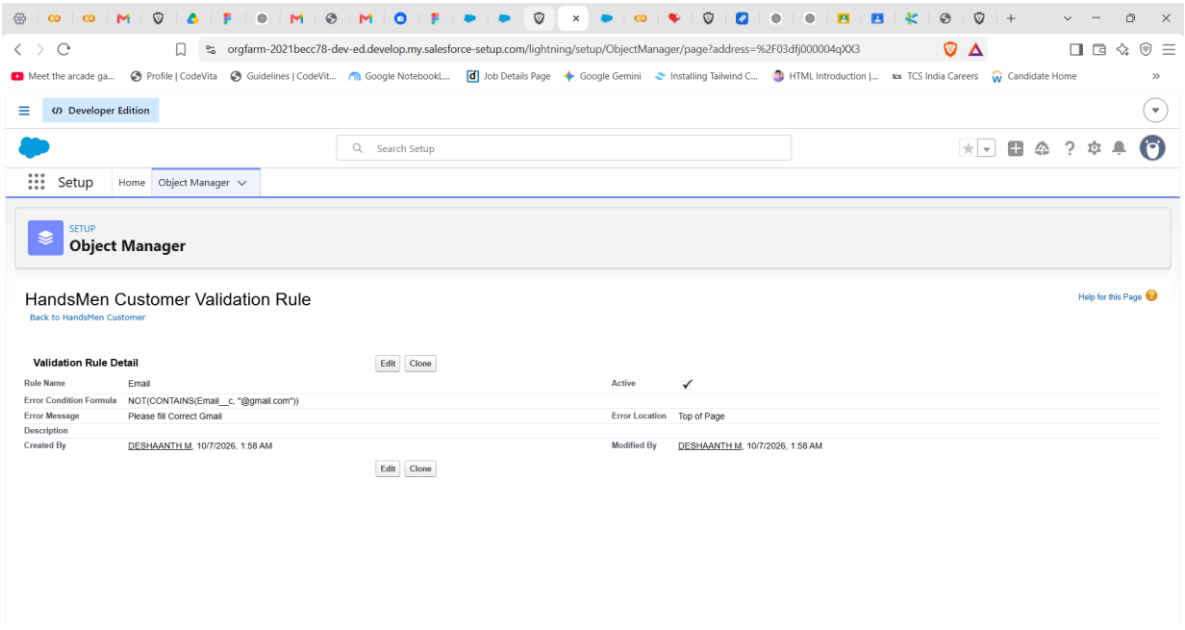


Figure: Customer email validation rule configured in Salesforce.

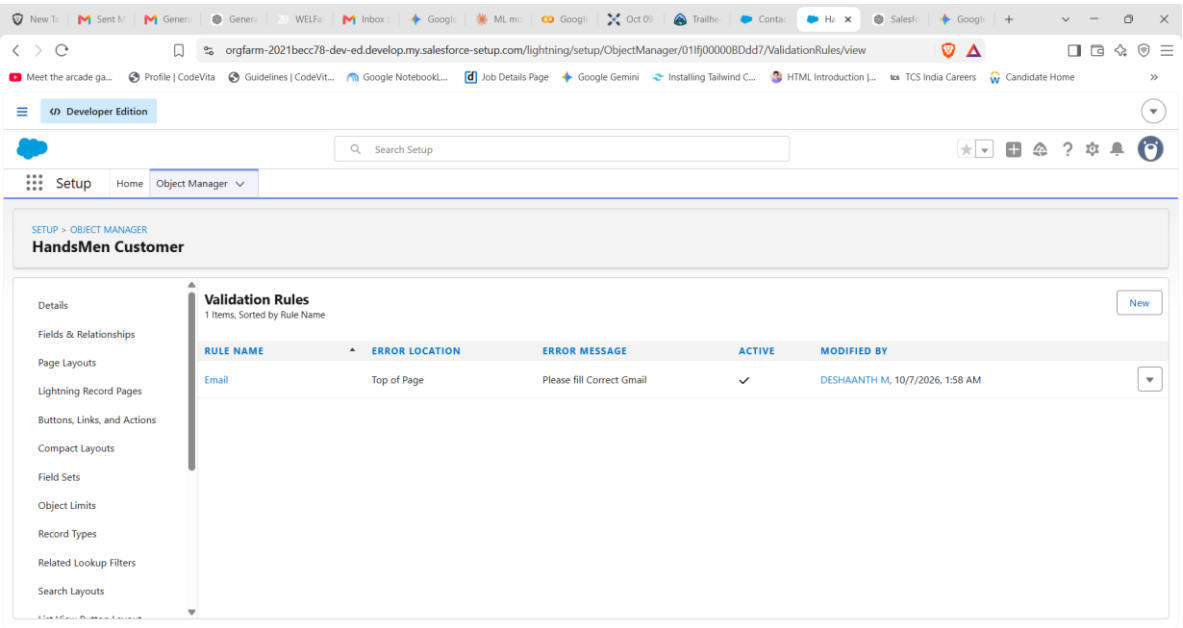


Figure: Active HandsMen Customer validation-rule list for Email.

Milestone 8: Email Templates

Definition: Classic email templates hold the reusable text for order confirmation and loyalty engagement emails.

Key steps:

- Setup > Classic Email Templates > New Template
- Choose the template type and folder
- Add the subject and body with merge fields
- Save and use the template in flow email alerts

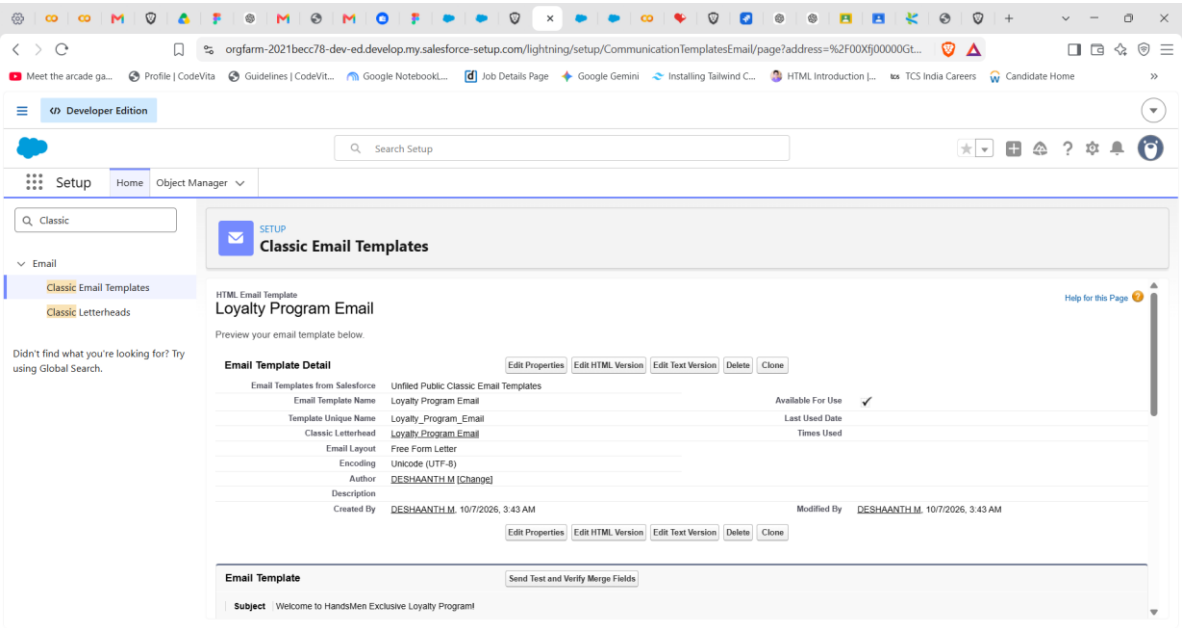


Figure: Loyalty Program Email template configuration.

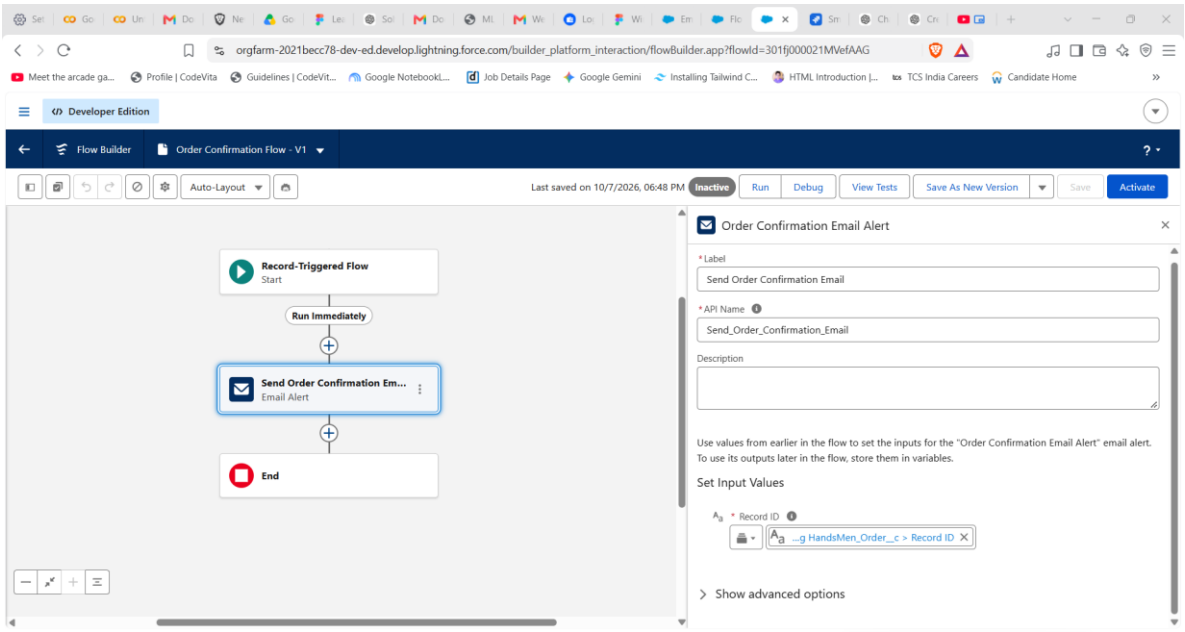


Figure: Order Confirmation Flow email-alert configuration.

Milestone 9: Flows (Automation)

Definition: Flow is Salesforce's declarative automation tool. Three flows handle order confirmation, low-stock alerts, and loyalty status updates.

Key steps:

- Setup > Flows > New Flow
- Choose Record-Triggered Flow for orders and inventory, and Scheduled-Triggered Flow for loyalty status
- Add decision, loop, update, and email-alert elements
- Save, activate, and run debug tests

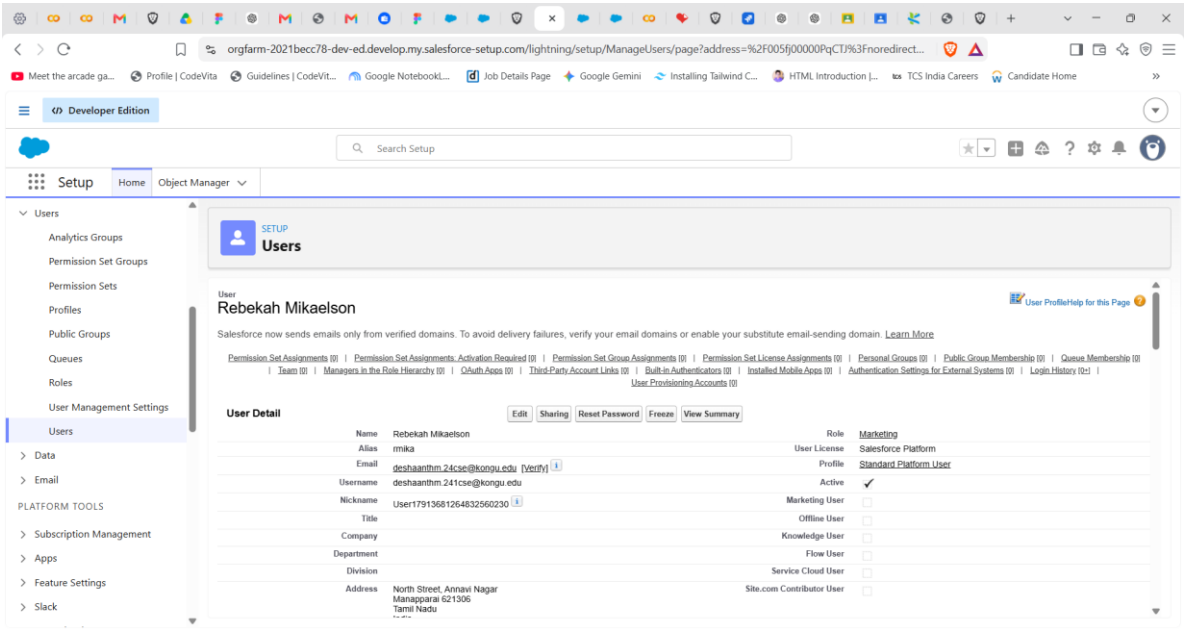


Figure: Order confirmation flow with an email alert action.

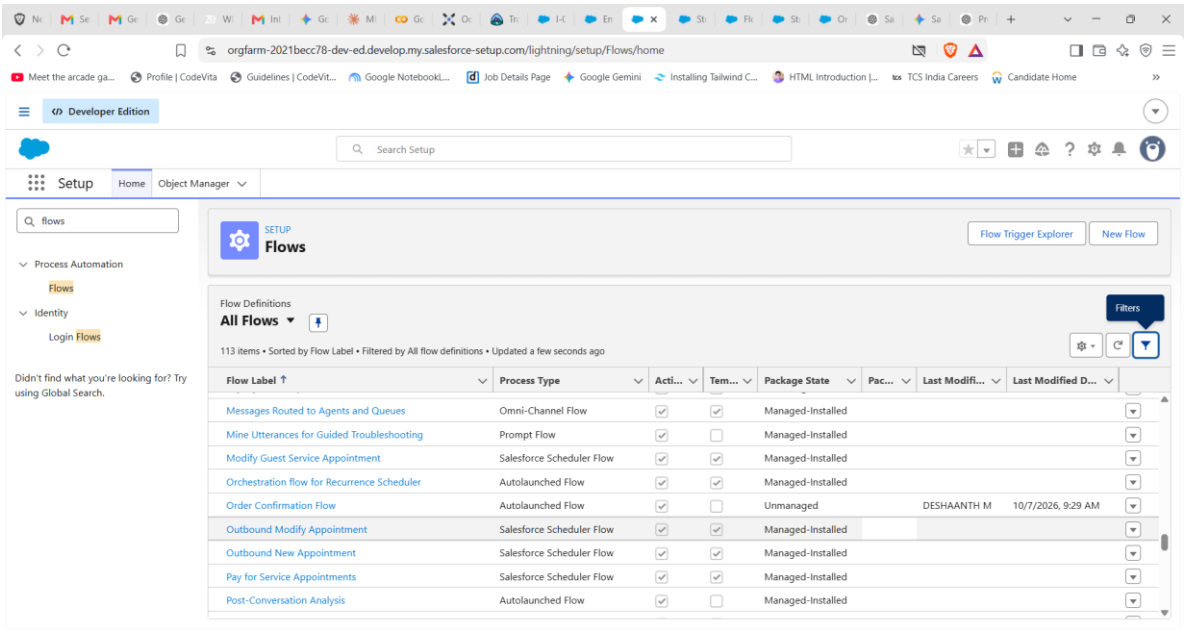


Figure: Salesforce flow list showing the Order Confirmation Flow as active.

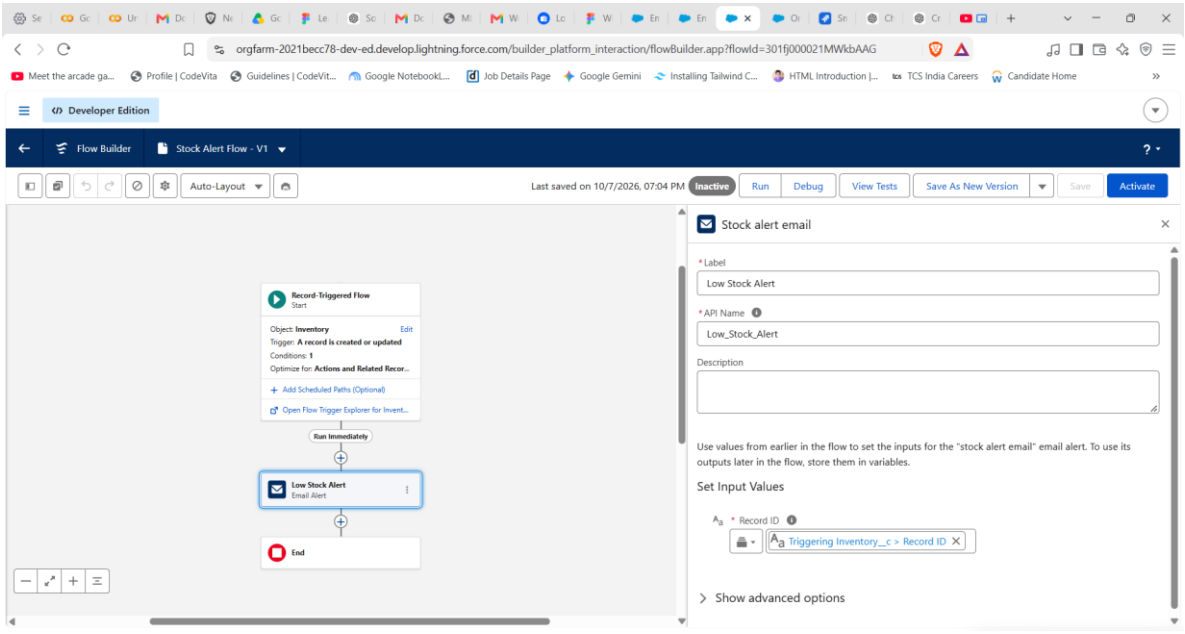


Figure: Low-stock alert flow with an email alert action.

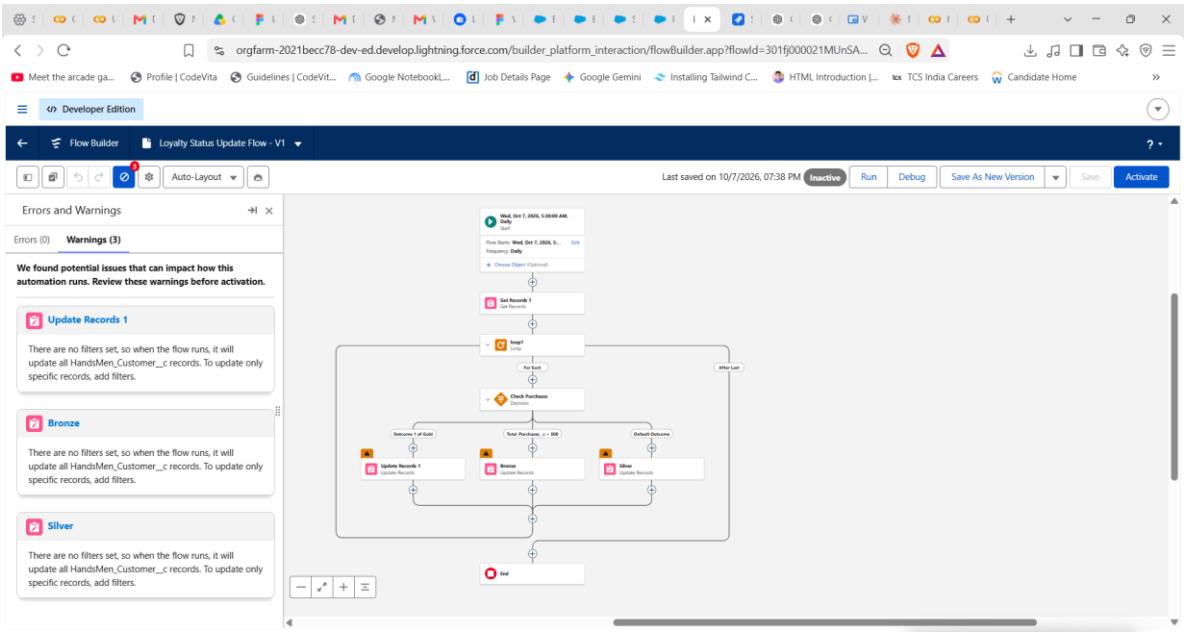


Figure: Scheduled loyalty status update flow and its decision paths.

Milestone 10: Apex Classes

Definition: Apex classes hold custom code. The org contains InventoryBatchJob and OrderTriggerHandler for batch and trigger-based logic.

Key steps:

- Setup > Apex Classes
- Review the InventoryBatchJob and OrderTriggerHandler classes
- Confirm they are listed and valid in the org

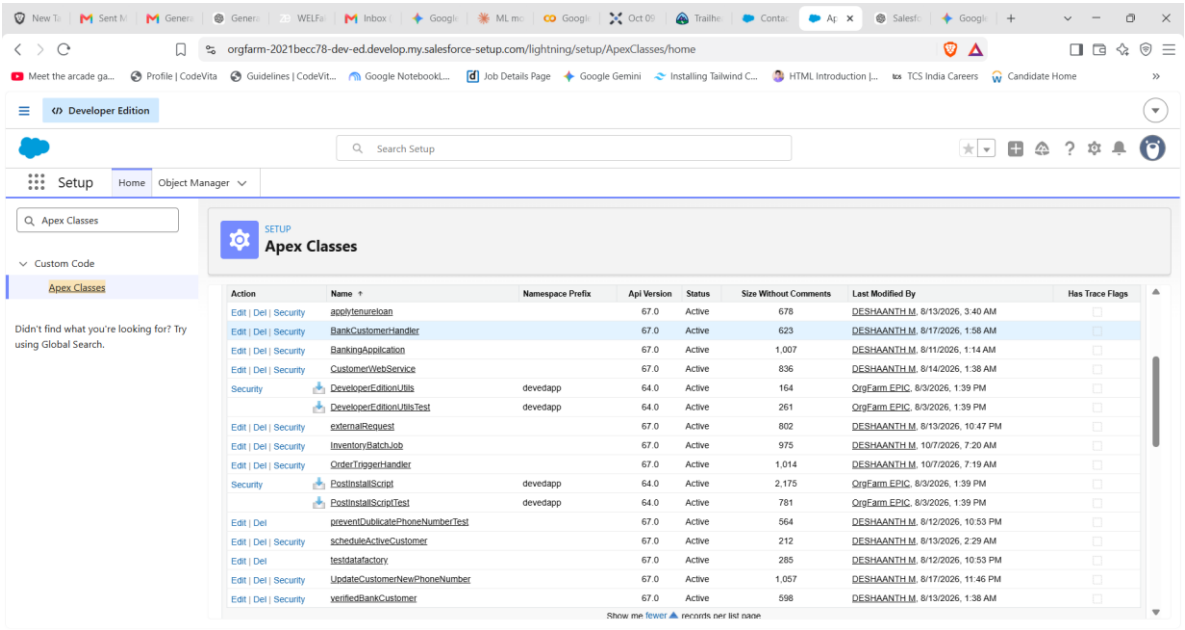


Figure: Apex Classes list showing InventoryBatchJob and OrderTriggerHandler among active classes.

Milestone 11: Security Setup

Definition: Security controls who can see and edit what. Roles, a profile, and a permission set give each team access to relevant data only.

Key steps:

- Create the Sales, Inventory, and Marketing roles
- Configure the Platform 1 profile and a permission set
- Assign one user to each role
- Test using Login As

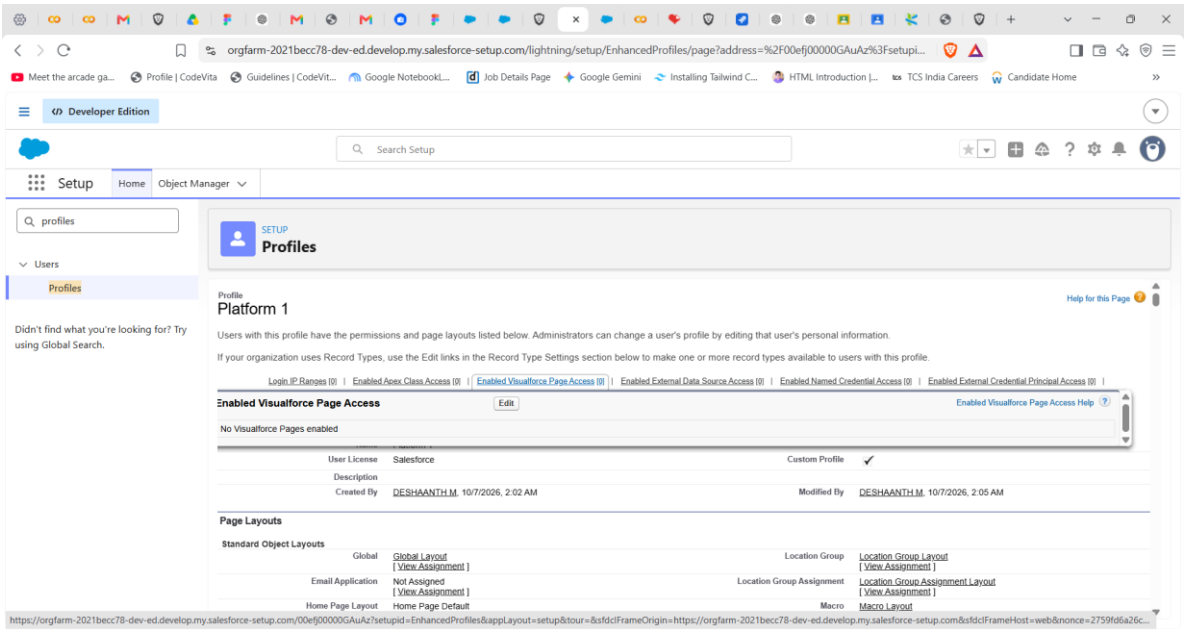


Figure: Platform 1 profile configuration.

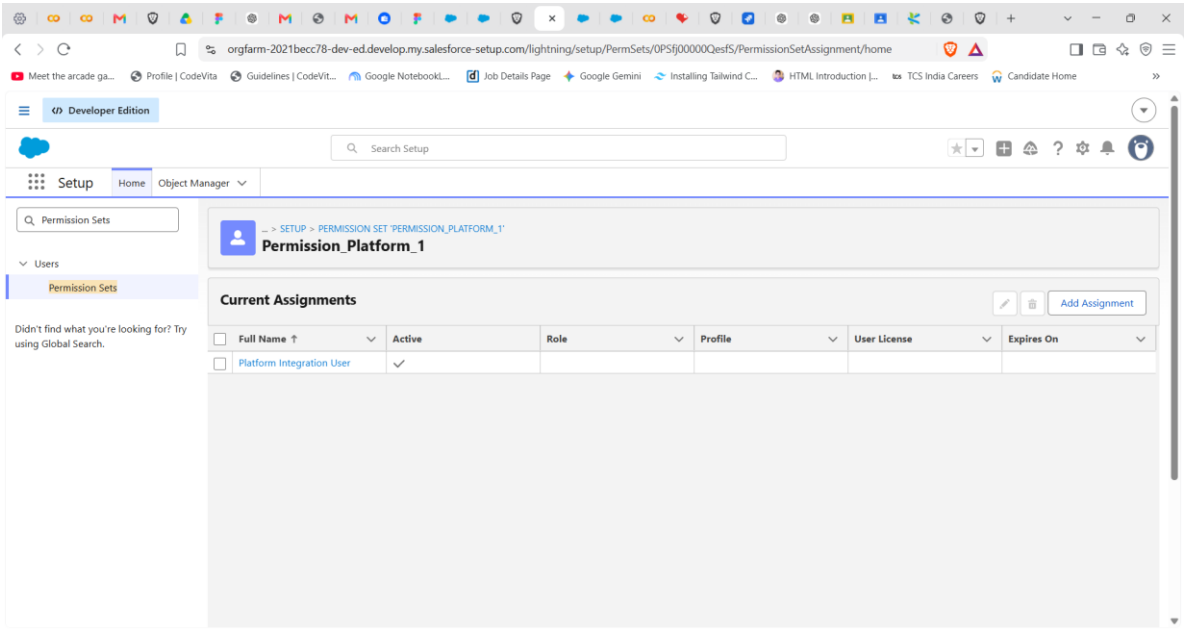


Figure: Permission Platform 1 permission-set assignment.

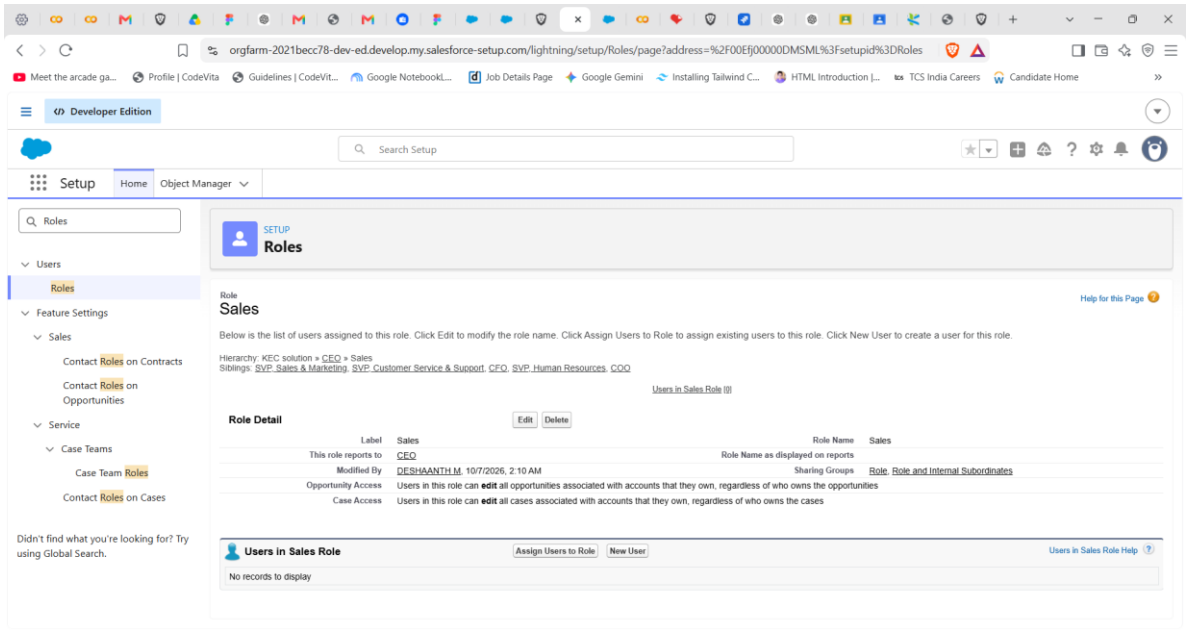


Figure: Sales role configuration.

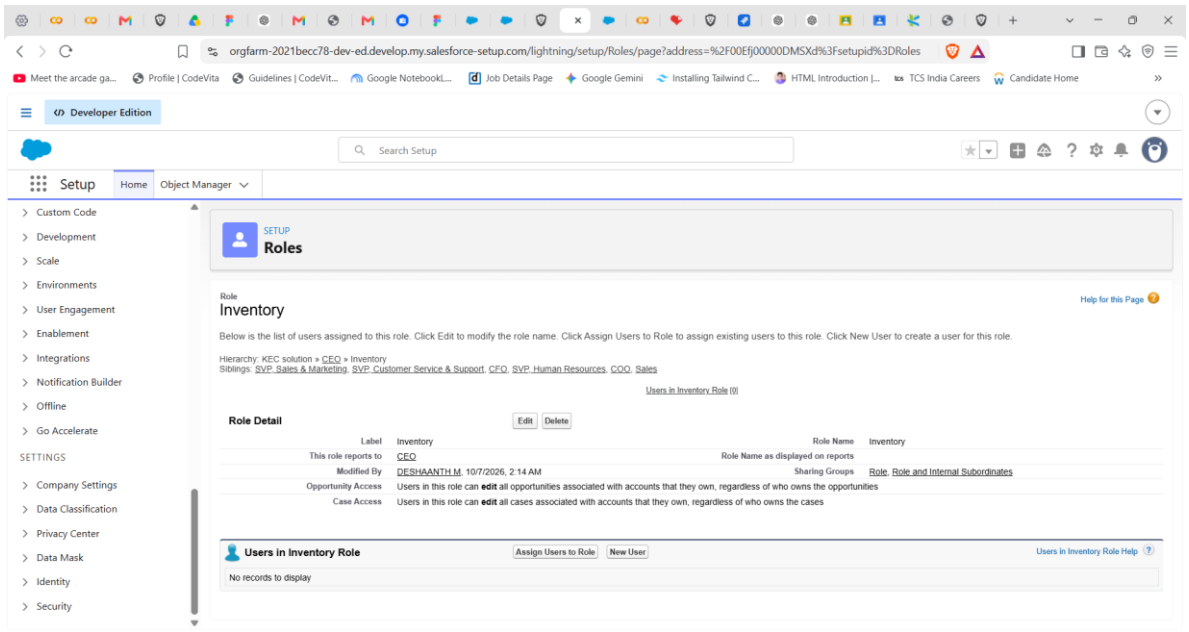


Figure: Inventory role configuration.

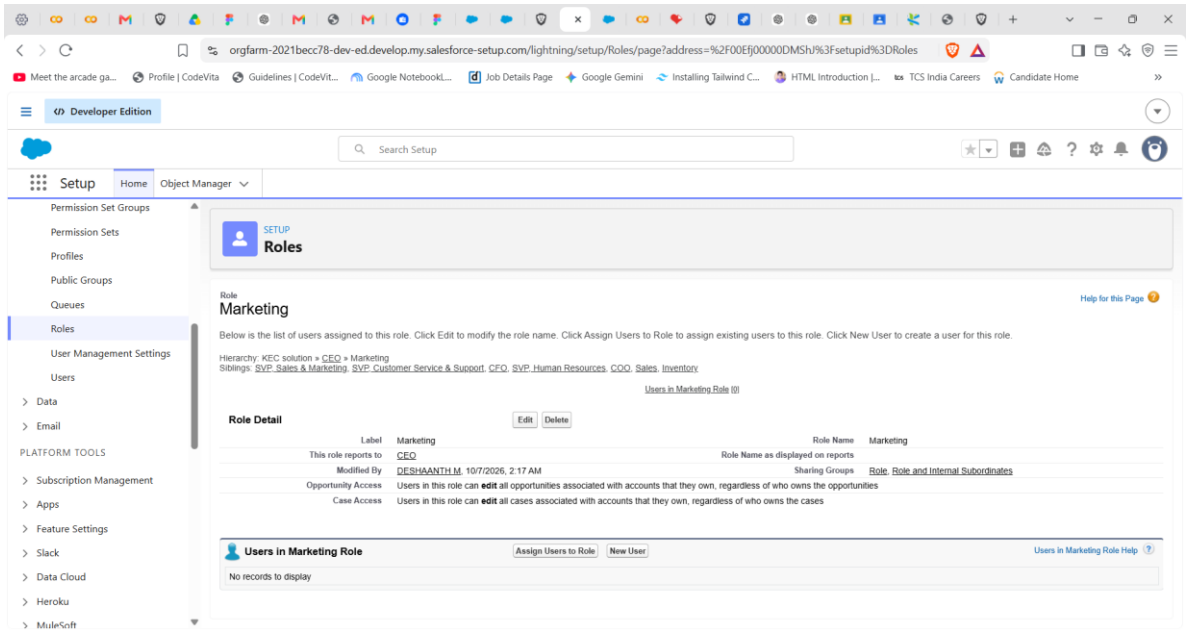


Figure: Marketing role configuration.

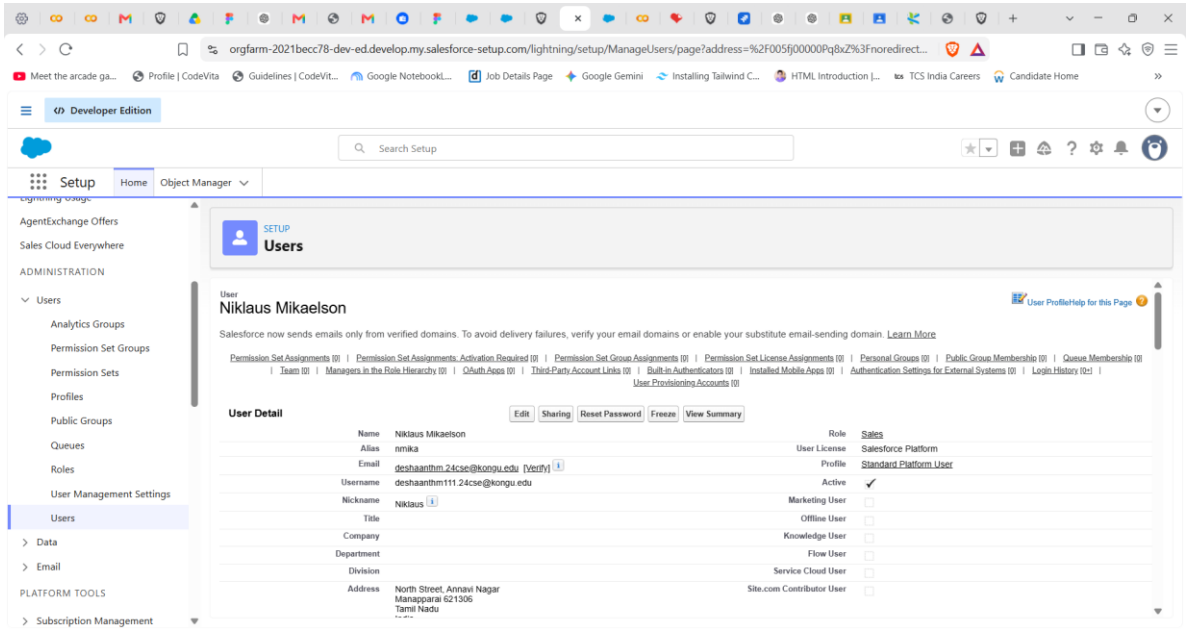


Figure: Sales user assigned to the Sales role.

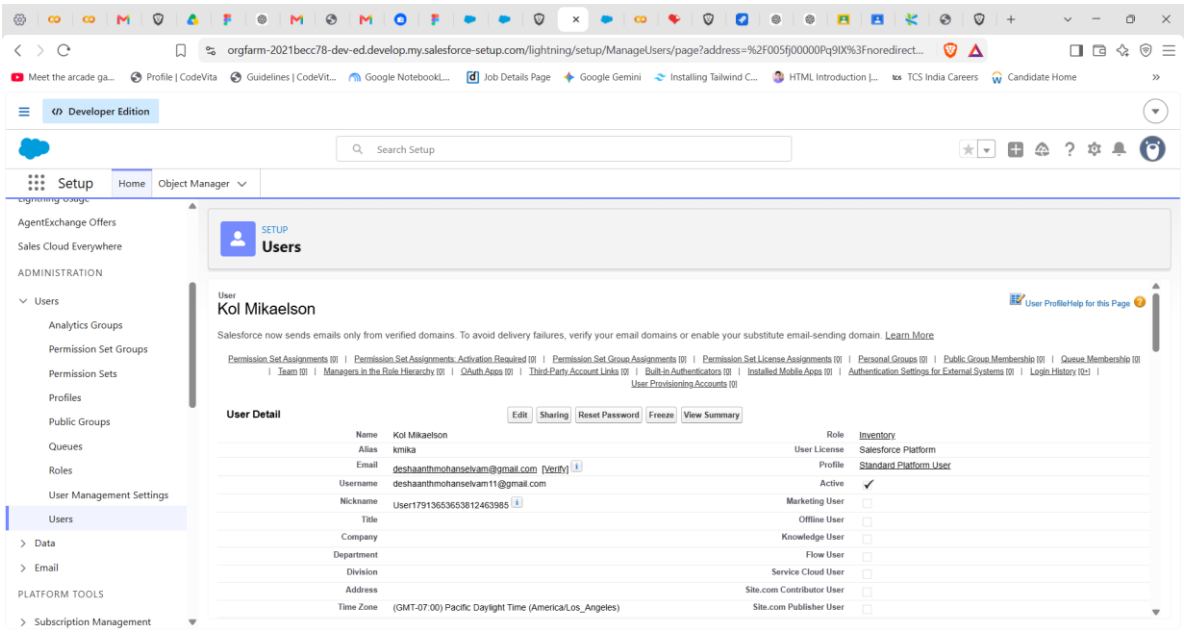


Figure: Inventory user assigned to the Inventory role.

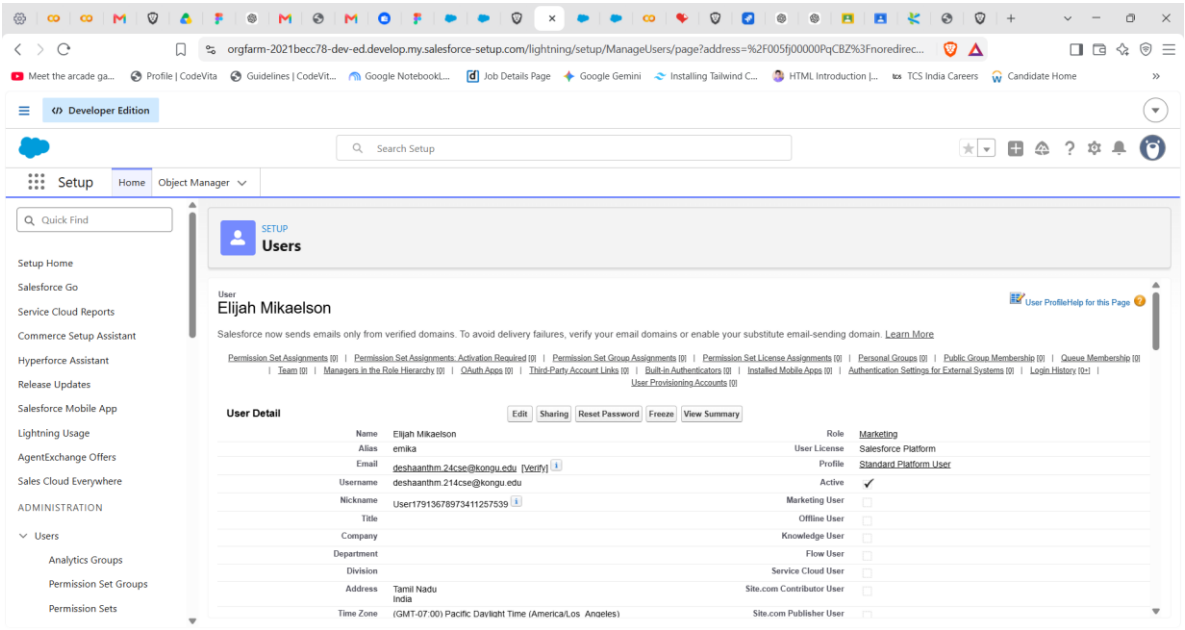


Figure: Marketing user assigned to the Marketing role.

Milestone 12: Testing and Verification

Definition: Testing confirms that validations and automation behave correctly using debug runs and sample records.

Key steps:

- Run debug tests for the Order Confirmation and Stock Alert flows
- Create sample customer, order, product, inventory, and campaign records
- Confirm the active flow list and Apex class list

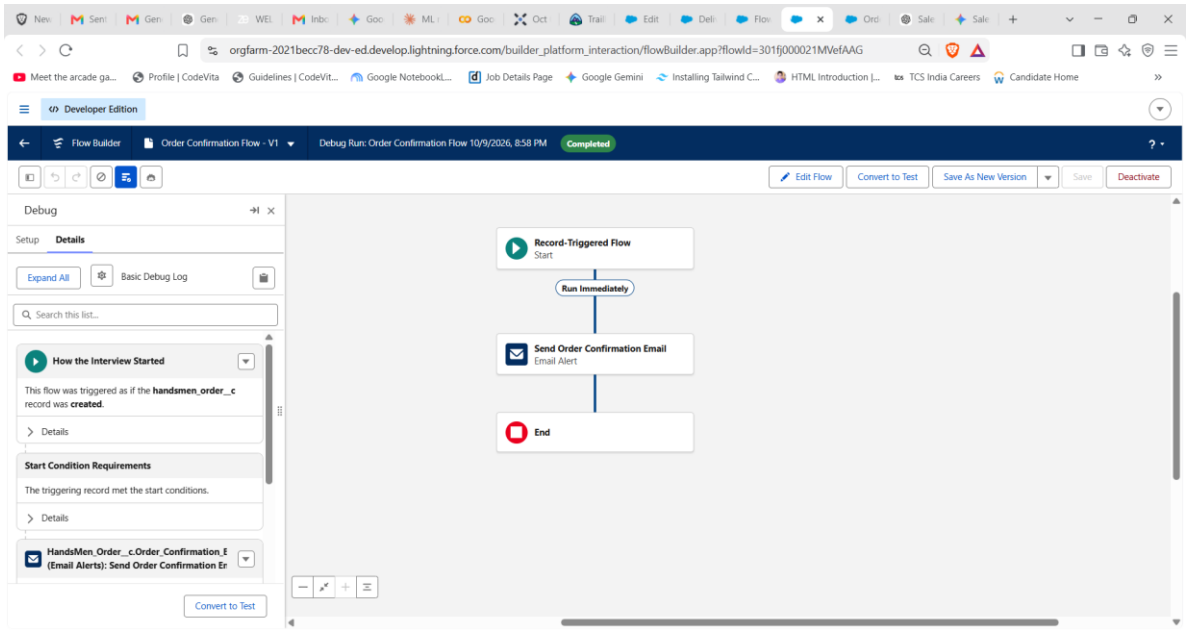


Figure: Completed debug run for the activated Order Confirmation Flow.

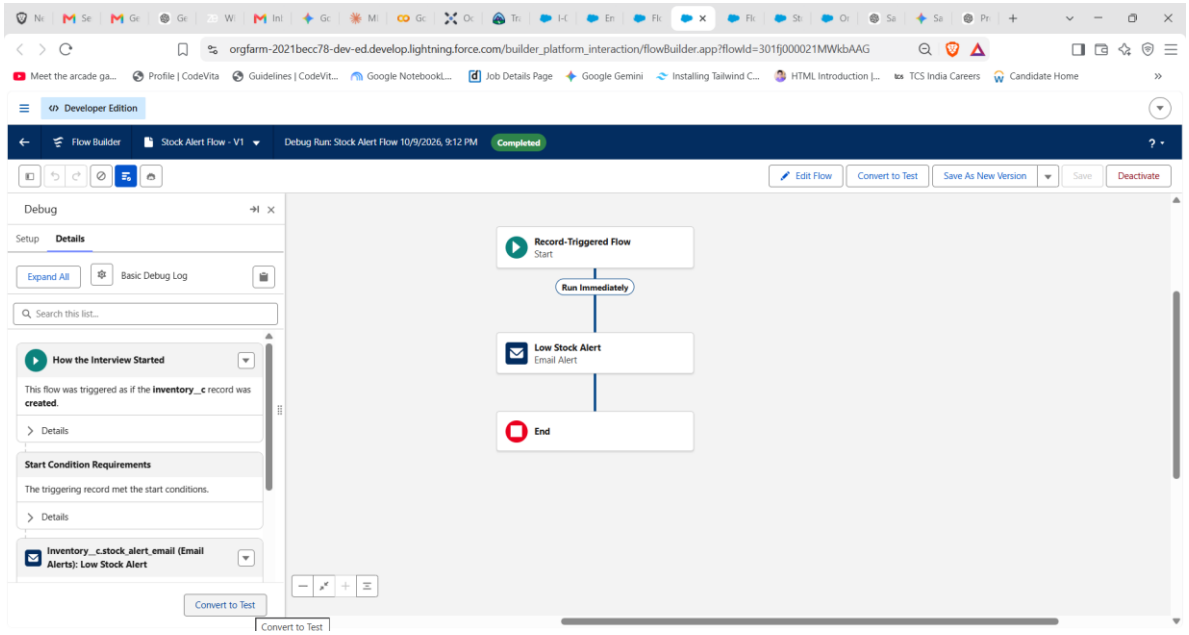


Figure: Completed debug run for the activated Stock Alert Flow.

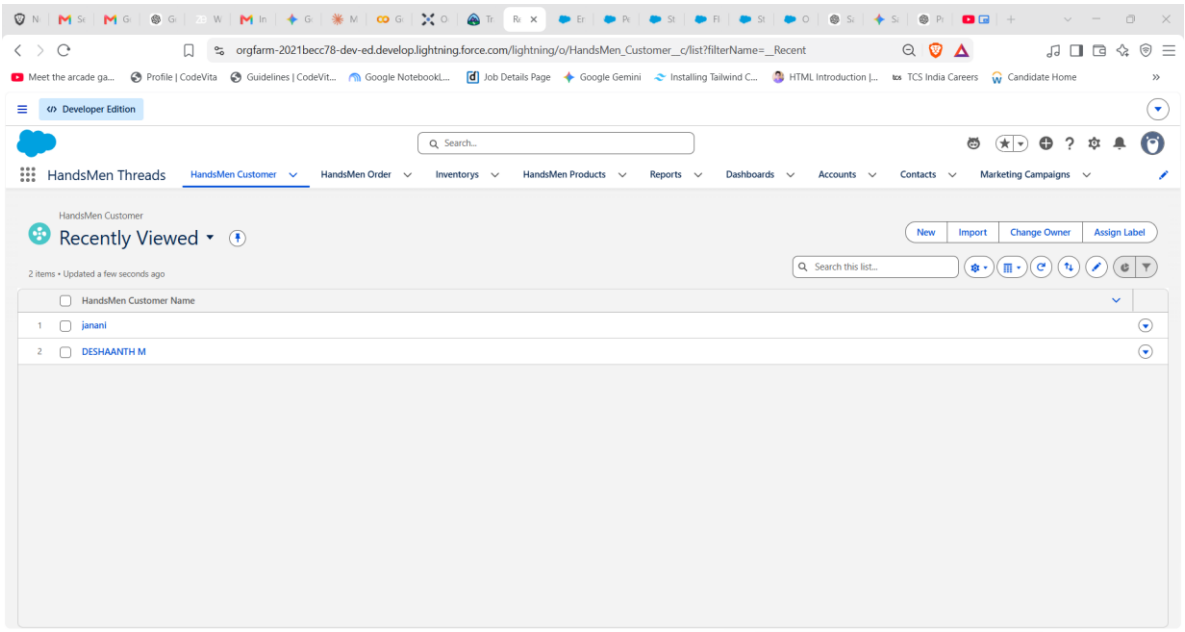


Figure: HandsMen Customer record list with two records.

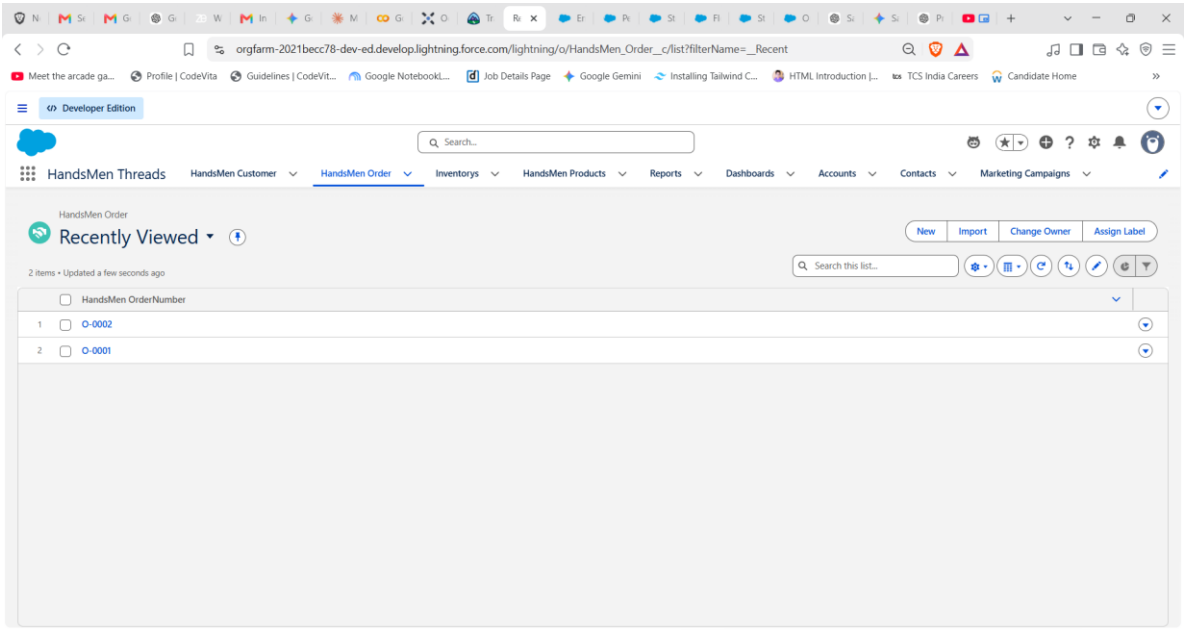


Figure: HandsMen Order record list with two orders.

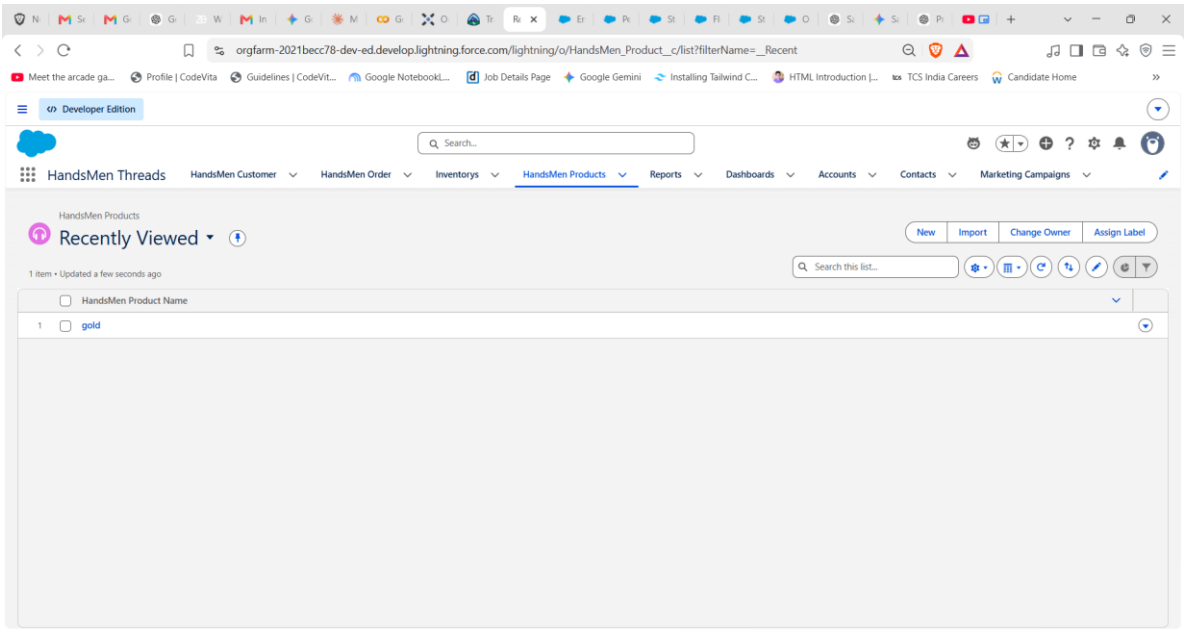


Figure: HandsMen Product record list with a product record.

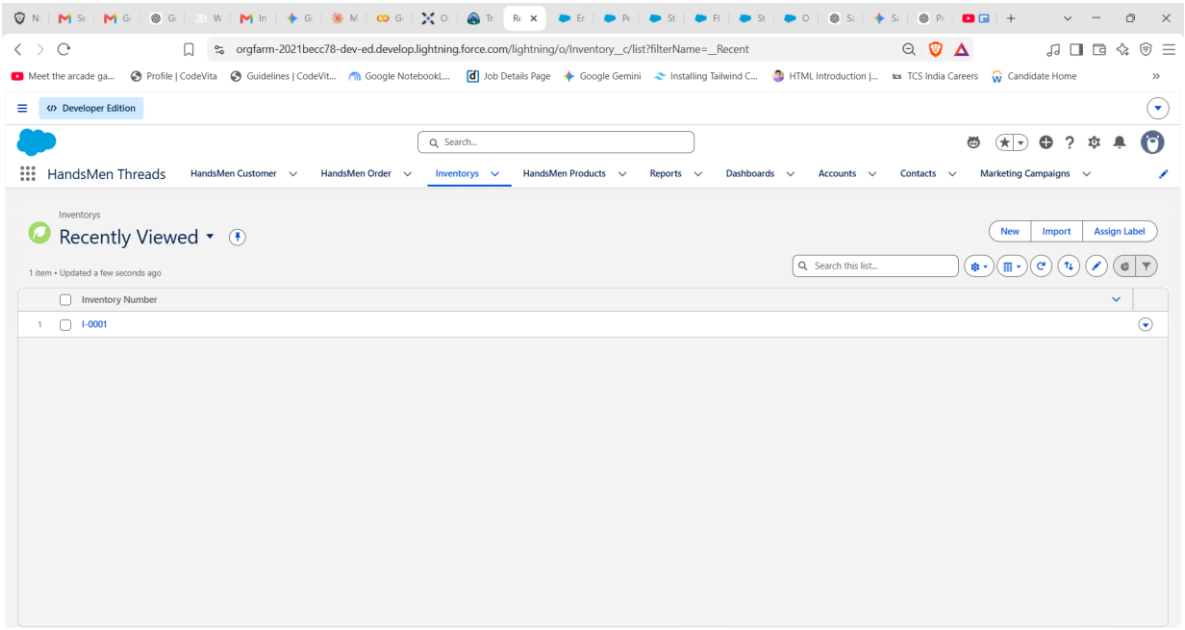


Figure: Inventory record list with an inventory record.

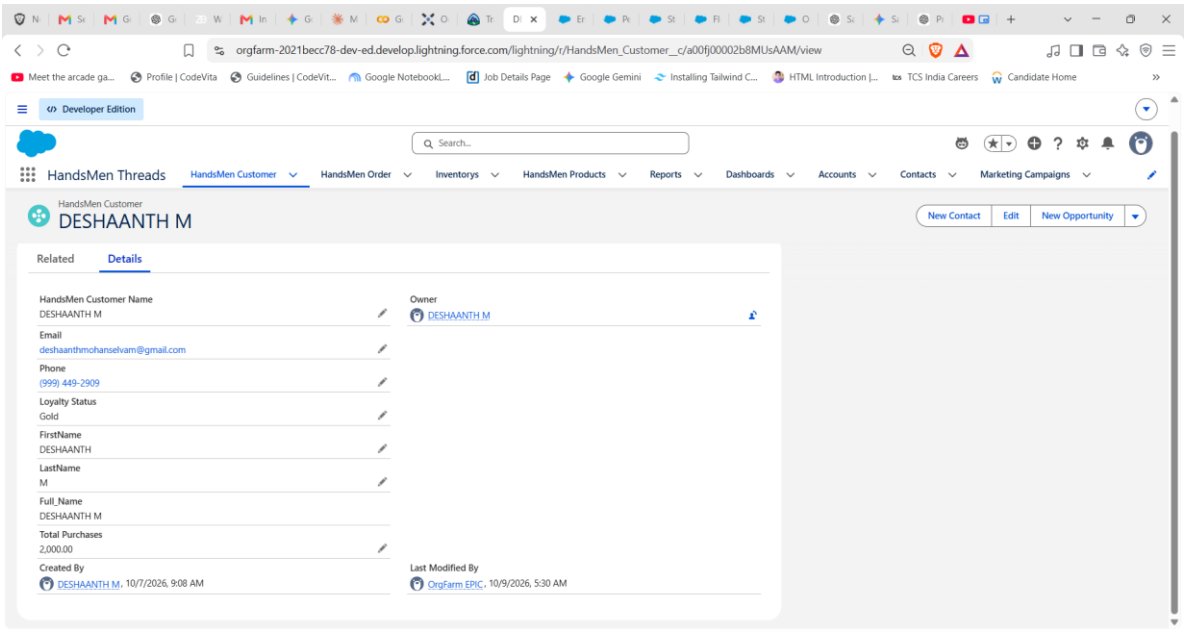


Figure: HandsMen Customer record with Gold loyalty status and total purchases.

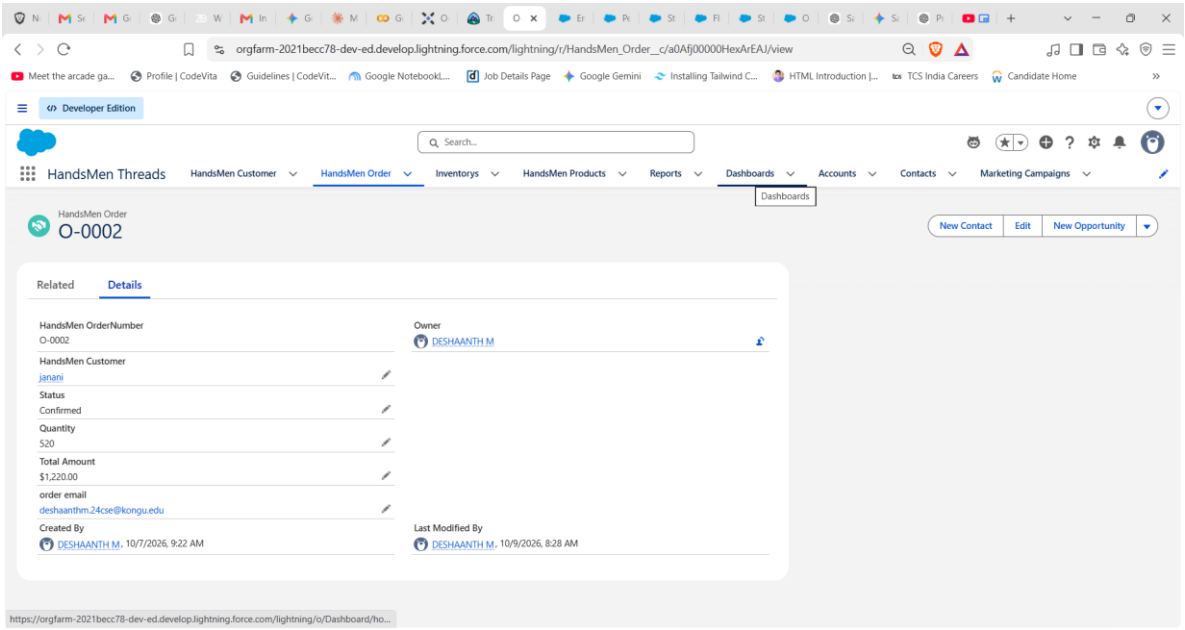


Figure: Confirmed order record O-0002 with quantity, total amount, customer, and email.

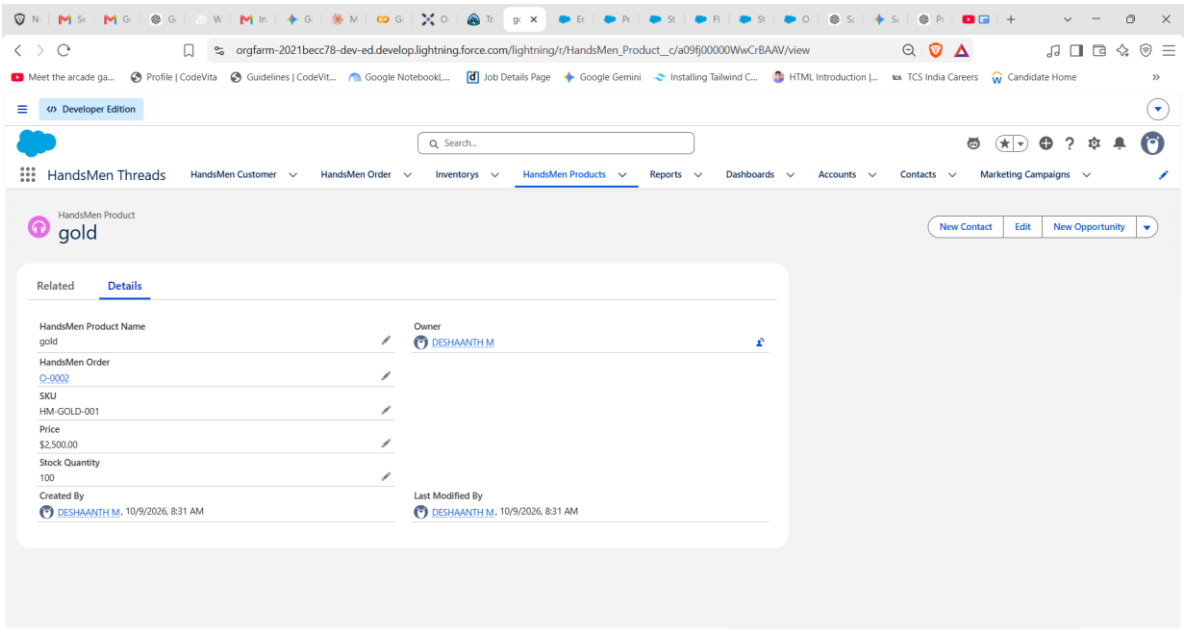


Figure: HandsMen Product record: SKU, price, stock quantity, and order relationship.

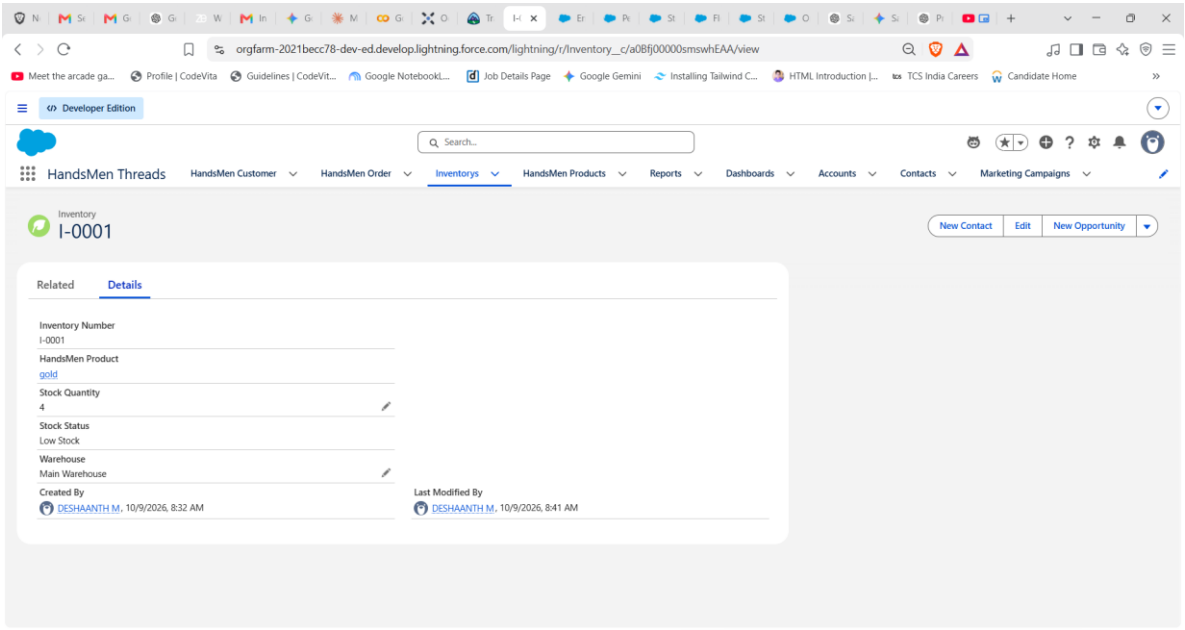


Figure: Inventory record I-0001: product, stock quantity, low-stock status, and warehouse.

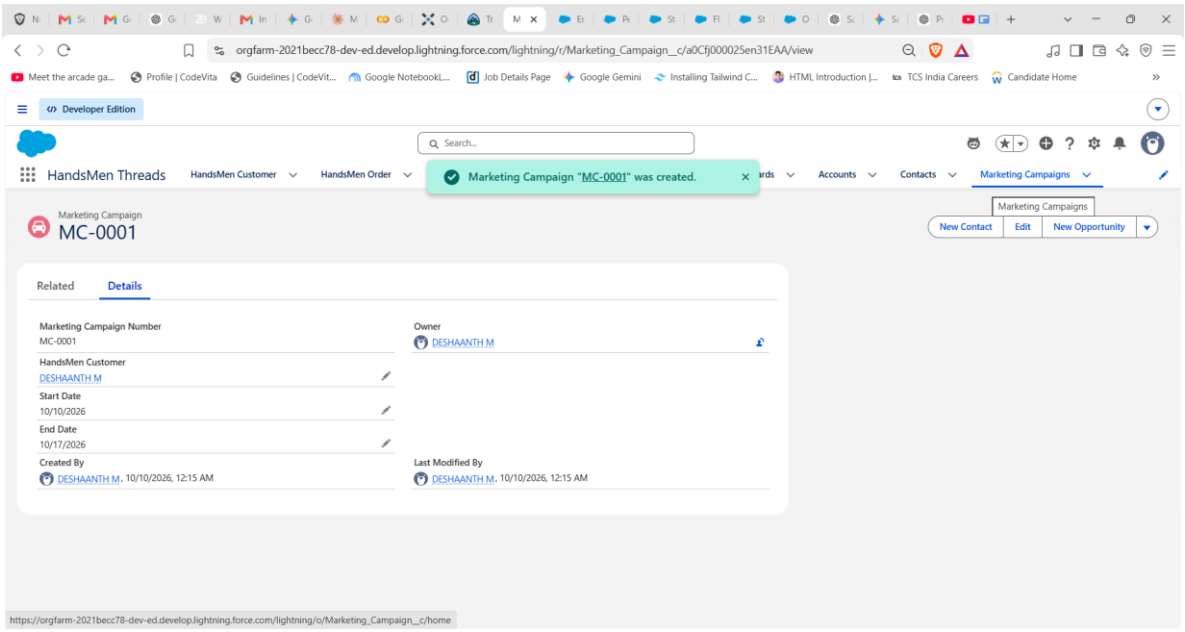


Figure: Marketing Campaign record MC-0001 linked to a customer and campaign dates.

7 Conclusion

HandsMen Threads provides a Salesforce-based foundation for mens fashion retail operations. The implemented configuration establishes dedicated custom objects, tabs, validation controls, role assignments, email templates, and flows for customer, order, stock, and loyalty activities. The application can be extended with reports, dashboards, approval processes, Lightning Web Components, and integrations as business requirements grow.

Outcomes and Benefits

- **Efficiency:** Less manual order follow-up and stock checking.
- **Accuracy:** Validation rules reduce invalid totals and stock values.
- **Speed:** Automatic confirmation emails and stock alerts shorten response times.
- **Customer engagement:** Loyalty status and campaign records support targeted marketing.
- **Accountability:** Records, flows, and role assignments are traceable in one system.
- **Scalability and maintainability:** The solution uses standard Salesforce configuration, so new products, warehouses, and users can be added without heavy custom code.

Future Enhancements

- Add sales, inventory, and campaign dashboards.
- Create Apex test classes and document coverage for the custom Apex implementation.
- Refine the loyalty scheduled flow so record updates target the loop item and use clear entry criteria.
- Add Lightning Web Components where declarative tools do not meet a future business requirement.
- Add an Experience Cloud portal for customer self-service order tracking.

Closing Statement

HandsMen Threads delivers on its core objectives: one platform, cleaner data, automatic customer communication, timely stock alerts, and loyalty-based marketing. Its configuration-first design gives the retailer a solid foundation to grow from.